

Amalgam Restorations



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INTRODUCTION

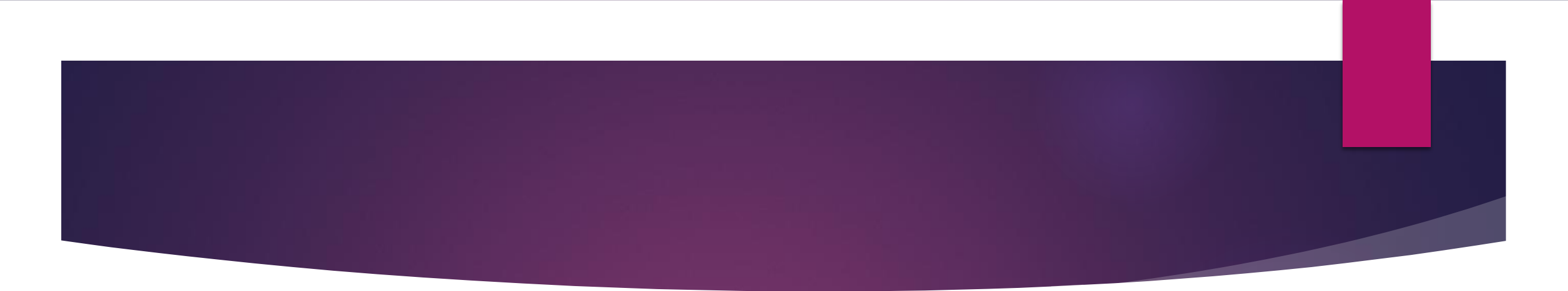
- ▶ The first form of silver mercury mixture was given by M Taveau in 1826 at Paris.
- ▶ Crawcour brothers in US introduced dental amalgam to the dentistry in 1833. Since 1850, dental amalgam has been used more than any other restorative material.
- ▶ The dental amalgam alloys consist of silver and other alloys like tin, copper and smaller amounts of zinc which are mixed with mercury.

Definition

- ▶ Alloy: is a union of two or more metals.
- ▶ Amalgam: is an alloy in which mercury occurs as a main constituent.
- ▶ Dental amalgam: is an alloy of mercury with silver, tin, and varying amounts of copper, zinc and other minor constituents.

INDICATIONS OF AMALGAM RESTORATION

1. Moderate to large class I, II preparation in which there is:
 - ▶ Heavy occlusion
 - ▶ Extension on the root surface
 - ▶ Problem of isolation.
2. Class V preparations in which:
 - ▶ Aesthetic is not a problem
 - ▶ Preparation entirely on root surface
 - ▶ Isolation is difficult.
3. Class VI preparations.
4. Class III preparations (sometimes) where isolation is difficult.



5. Used as a foundation

- ▶ In cases of grossly decayed teeth while planning for cast restoration.

6. Used as a post endodontic restoration.

7. Teeth having non definitive pulpal prognosis

- ▶ used as type of interim restoration before assessment of pulpal status of the tooth.

8. Tooth having fractured cusp can be restored with the help of amalgam using pin and slot.

CONTRAINDICATIONS OF AMALGAM RESTORATION

1. Esthetics:

- ▶ Use of amalgam is avoided in aesthetic areas of oral cavity. So, preparations class III, IV, V usually are not indicated except in certain cases.

2. Small to moderate class I and class II preparations

- ▶ should be restored with composite rather than amalgam as former results in more conservative tooth preparation.

3. Extensive loss of tooth structure.

4. Restoration of primary tooth

5. Restoration of periodontally weak tooth.

ADVANTAGES OF SILVER AMALGAM

- ▶ Favorable long-term clinical results
- ▶ Ease of manipulation/ use.
- ▶ High compressive strength.
- ▶ Wider range of application.
- ▶ Physical characteristics of amalgam are comparable to enamel and dentin.
- ▶ Less technique sensitive.
- ▶ Self-sealing ability, satisfactory marginal adaptation- reduce microleakage.
- ▶ Biocompatible.
- ▶ Excellent wear resistance.
- ▶ Economic/Low cost.
- ▶ Bonded to tooth structure (Bonded amalgam).

DISADVANTAGES OF SILVER AMALGAM

- ▶ Less aesthetic.
- ▶ Less conservative (extensive preparation).
- ▶ Corrosion and tarnish - causing discoloration.
- ▶ Does not bond to tooth.
- ▶ Non-insulating.
- ▶ Marginal degradation (low copper alloys).
- ▶ Lack of reinforcement of weakened tooth structure.
- ▶ Poor tensile strength making it a brittle material.
- ▶ Difficulty in restoring proper tooth anatomy.
- ▶ Galvanic current shock

CLASSIFICATION OF AMALGAM

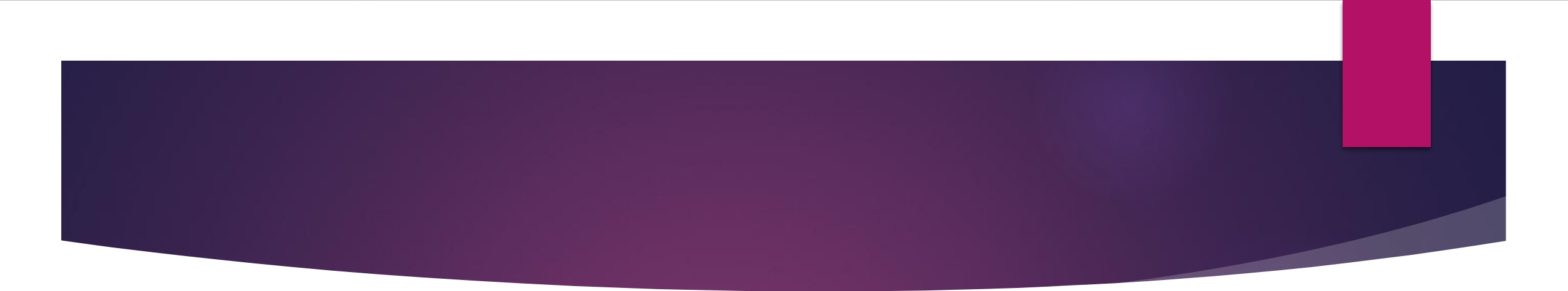
- ▶ There are different ways of classifying amalgam alloys. These are:

1. Based on shape of particles

- ▶ a. Irregular: shape of particles is irregular, may be in the shape of spindles or shavings.
- ▶ b. Spherical: shape of particle is spherical with smooth surface.
- ▶ c. Spheroidal: shape of particle is spheroidal with irregular surface.

2. Based on copper content

- ▶ a. Low copper alloy: Copper is in range of 2-6%
- ▶ b. High copper alloy: Copper is in range of 6-30%
 - Admixed alloy
 - Single composition alloy



3. Based on zinc content

- ▶ a. Zinc-containing alloys: Zinc is in range of >0.01
- ▶ b. Zinc-free alloys: Zinc is in range of $< 0.01\%$

4. Based on the number of alloy metals

- ▶ a. Binary alloys: Contain 2 metals, i.e. silver and tin.
- ▶ b. Ternary alloys: Contain 3 metals, i.e. silver, tin and copper.
- ▶ c. Quaternary alloys: Contain 4 metals, i.e. silver, tin, copper and zinc.

Out of these, quaternary alloys are most acceptable.

5. Based on presence of noble metals

- ▶ a. noble metal alloy(palladium, platinum, gold)
- ▶ b. non-noble metal alloys

6. Based on the generation

- ▶ Class I. Silver and tin in ratio(8:1)
- ▶ Class II. Silver , tin, copper and zinc, cu(4%)
- ▶ Class III. Silver eutectic alloy added to original alloy
- ▶ Class IV. Copper content increased to 29 percent
- ▶ Class V. Indium added to mixture of silver, tin and copper
- ▶ Class VI. Noble metals added such as palladium

COMPOSITION OF AMALGAM POWDER

- ▶ Amalgam alloy- is composed of silver-tin alloy with varying amounts of copper, zinc, indium and palladium.
- ▶ Dental amalgam alloys are mainly of two types: low copper and high copper alloys.

A. Low copper alloys (contain up to 6% by weight of copper.)

- ▶ Silver.....63-70%
- ▶ Tin.....26-28%
- ▶ Copper.....2-6%
- ▶ Zinc.....0-2%

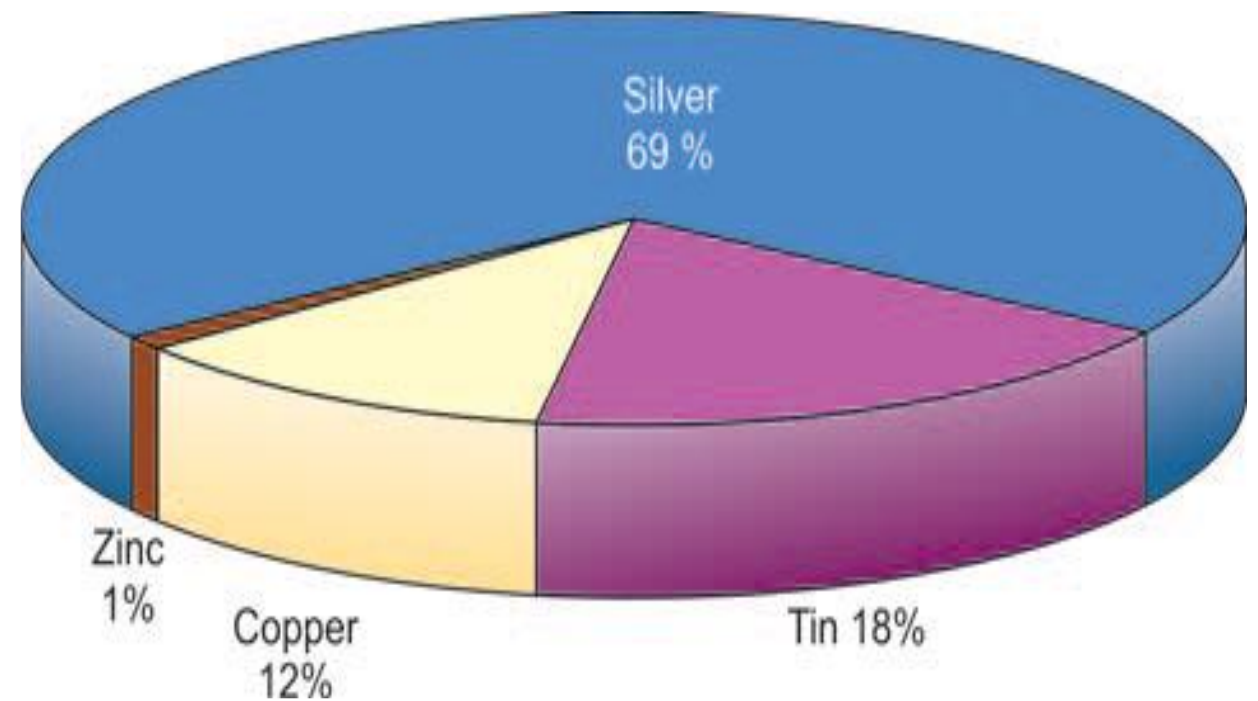
► B. High copper alloys (copper content lies between 6-30%).

► Silver.....69%

► Tin.....17-18%

► Copper.....12-13%

► Zinc.....1%



NOTE

- ▶ In general, amalgam alloy consists of:
 - ▶ Silver 40% minimum,
 - ▶ Tin 32% maximum,
 - ▶ Copper 30% maximum,
 - ▶ Zinc 2% maximum, and
 - ▶ Sometimes traces of indium or palladium.
- ▶ In preamalgamated alloys, mercury 3% is used which react more rapidly when mixed with mercury. Mercury used for dental amalgam is purified by distillation.

Percentage of elements by weight

<i>Alloy</i>	<i>Particle</i>	<i>Silver</i>	<i>Tin</i>	<i>Copper</i>	<i>Zinc</i>	<i>Palladium</i>	<i>Indium</i>
• Low copper	Lathecut or Spherical	65 to 77	26 to 28	2 to 5	0 to 2	0	0
• High copper							
– Admixed	Lathecut	40 to 70	26 to 30	13 to 30	0 to 1	0	0
	Spherical	40 to 70	0 to 30	20 to 30	0	0	0
– Unicompositional	Spherical	40 to 60	22 to 30	15 to 30	0 to 1	0 to 4	0

Effects of Constituent Metals on the Properties of Amalgam

► **Silver:**

- It has the following effects on the properties of amalgam.
 - Increases strength
 - Increases setting expansion
 - Reduces setting time
 - Resists tarnish and corrosion
 - Decreases flow.

► **Tin:**

- Tin helps in formation of a silver/tin compound (Ag_3Sn). This is the gamma phase which readily undergoes an amalgamation reaction with mercury. Tin causes following effects:
 - Increases setting time
 - Reduces strength, hardness, and setting expansion.



► **Copper:**

- Reduces tarnish and corrosion
- Reduces creep
- Strengthening effect on the set amalgam
- Helps in uniform comminution of the alloy.

► **Zinc:**

- Its presence is not essential. It may vary from 0 to 2% by weight. It has the following effects on the properties of amalgam:
- Scavenges the available oxygen to impede oxidization of Ag, Sn or Cu during alloy ingot manufacture.
- If zinc-containing alloys are contaminated with moisture, Zn gives rise to delayed or secondary expansion.

► **Palladium** (0 to 1% by weight)

- Improves the corrosion resistance and the mechanical properties.

► **Indium** (0 to 4% by weight)

- It decreases the evaporation of mercury and the amount of mercury required to wet the alloy particles.

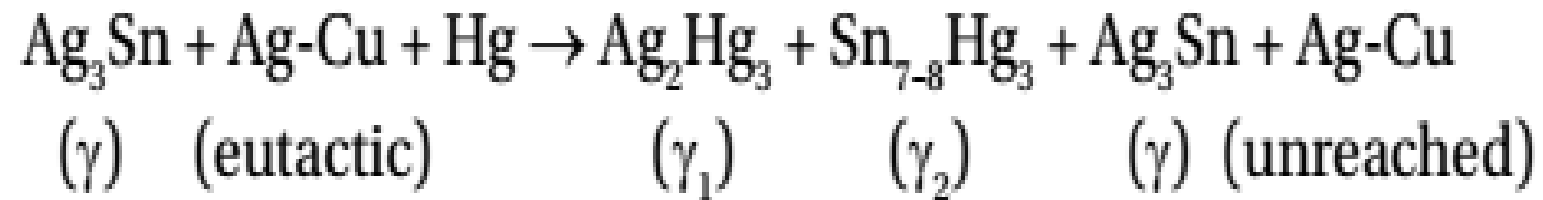
Proportioning

- Usually Alloy/mercury ratios range between 5 : 8 and 10 : 8. But to achieve optimum properties of the amalgam, mercury should be less than 50%. For lathe cut alloys, it is 45%. For spherical alloys, it is 40% Hg.

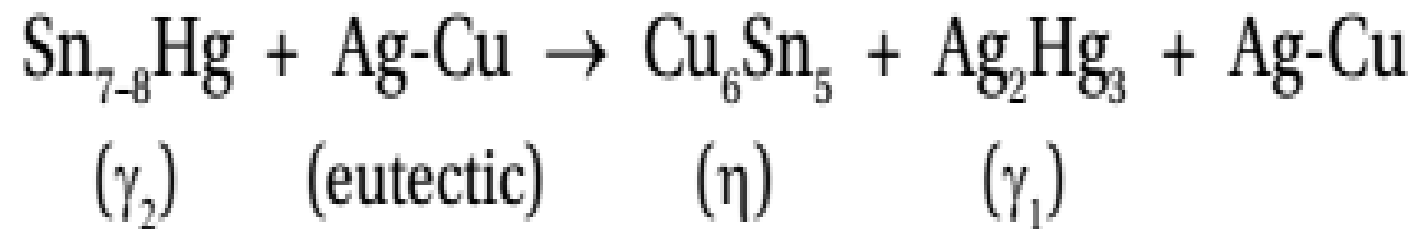
Setting Reaction of Amalgam

- ▶ For Lathe-cut Low Copper Alloys
- ▶ Mercury reacts with alloy particles to form two products, i.e. the silver mercury phase and tin mercury phase. After this reaction, the unreacted particles are embedded in the matrix of reaction products with mercury. The reaction is as following:
- ▶
$$\text{Ag}_3\text{Sn} + \text{Hg} \rightarrow \text{Ag}_2\text{Hg}_3 + \text{Sn}_8\text{Hg} + \text{Ag}_3\text{Sn}$$
- ▶
$$\begin{matrix} (\gamma) & & (\gamma_1) & & (\gamma_2) & & (\gamma) \end{matrix}$$
- ▶ In lathe-cut low copper amalgams both γ_1 and γ_2 form a continuous network. Since γ_2 phase is least corrosion resistant phase, its distribution in reaction product is important.

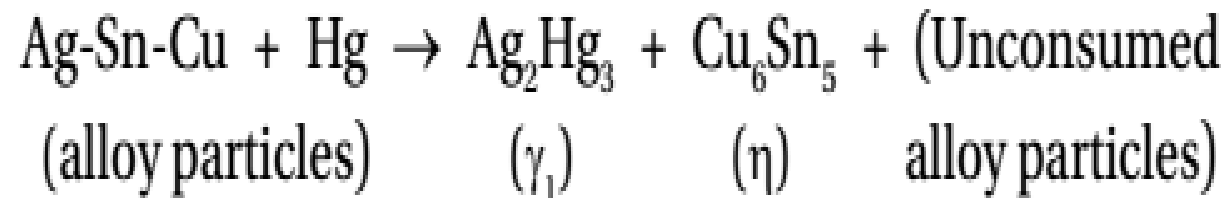
- ▶ For Admixed High Copper Alloys
- ▶ For high copper alloys, the reaction is different. It occurs in two phases.
- ▶ a. Initial reaction : is similar to that of low copper alloys, i.e.



- ▶ b. Second phase reaction: involves the silver copper phase (Ag-Cu).
- ▶ It reacts with γ_2 (Sn₈Hg) to form Ag₂Hg₃ and Cu₆Sn₅ phase. The mercury released from Sn₈Hg (γ_2 phase) reacts with silver to form Ag₂Hg₃ (γ_1) phase.
- ▶ This reaction goes on. After one week, the γ_2 phase reacts completely with eutectic and replace all the γ_2 phase by γ and γ_1 phase.



- ▶ For Unicompositional Silver Alloy
- ▶ The difference in admix type and the unicompositional alloys is that in latter the eutectic phase, i.e. Ag-Cu phase is absent and the reaction is directly with silver, copper and tin phases. In these only silver reacts with mercury and the tin remains bound to copper.
- ▶ The final phase formed is Cu₆Sn₅ (η) phase and there is no Sn₈Hg₃ (γ₂) phase.



Phases of silver amalgam

<i>Code</i>	<i>Component</i>
(γ) gamma	Ag ₃ Sn (Silver Tin phase)
(γ_1) gamma 1	Ag ₂ Hg ₃ (Silver Mercury phase)
(γ_2) gamma 2	Sn ₇₋₈ Hg ₃ (Tin Mercury phase)
(ϵ) epsilon	Cu ₃ Sn (Copper Tin phase)
(η) eta	Cu ₆ Sn ₅ (Copper Tin phase)

Difference between High Copper and Low Copper Alloys

	<i>High copper alloys</i>	<i>Low copper alloys</i>
Copper content	6-30%	<6%
Mercury required for amalgamation	Less	More
Setting reaction	Fast setting	Slow setting
Amalgamation speed and energy	Require high-speed and energy for amalgamation since copper has low solubility in mercury	Require less speed and low energy for amalgamation
Dormant phase	It is Cu_6Sn_5 , i.e. η phase	It is Ag_2Hg_3 , i.e. η_1 phase
Tarnish and corrosion	It is due to copper rich phase, i.e. Cu_6Sn_5 .	It is due to gamma 2 phase, i.e. $\text{Sn}_{7-8}\text{Hg}_3$
Creep	Less creep (<1%)	More creep (1-8%)
Compressive strength	High (250-500 Mpa)	Low (150-350 Mpa)
Dimensional change	Less (1-9 $\mu\text{m}/\text{cm}$)	More (10-20 $\mu\text{m}/\text{cm}$)

PHYSICAL PROPERTIES OF AMALGAM

DIMENSIONAL CHANGE

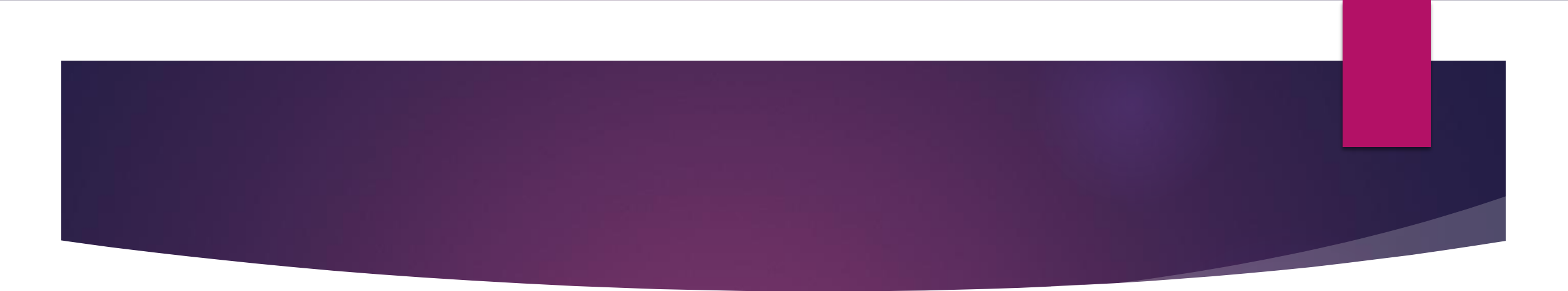
- ▶ Small amount of contraction occurs in the first half an hour after trituration because mercury diffuses into the silver and tin and the mix dissolves in the mercury.

▶ **Factors Affecting Dimensional Changes of Amalgam**

1. **Type of alloy being used**

- ▶ For example, single composition spherical alloys contract more than single composition lathe cut or admixed alloys.

2. **Condensation technique**, i.e. more mercury removed from the alloy, the more it will contract.



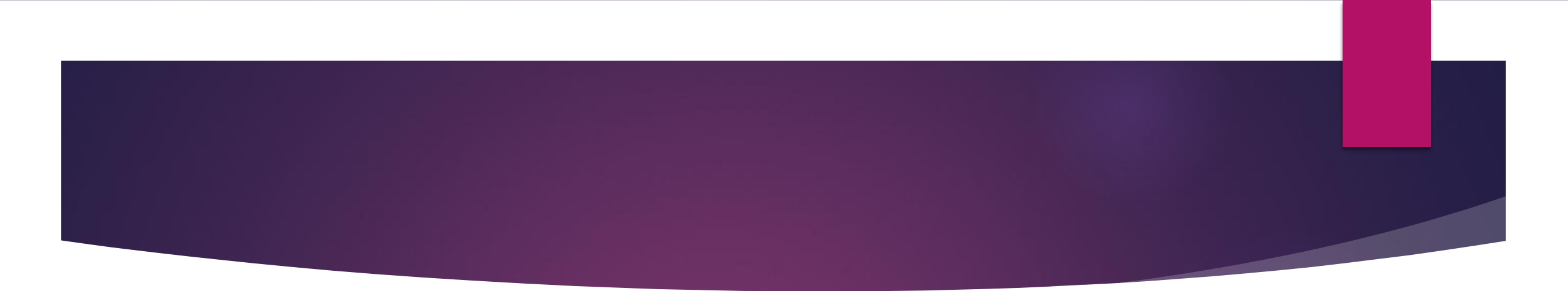
3. **Trituration time**, over trituration causes contraction.

4. **Presence of zinc:**

- ▶ If zinc containing amalgam comes in contact with moisture or saliva during condensation or trituration, it can result in delayed expansion after 3 to 5 days of restoration.
- ▶ This occurs due to formation of zinc oxide and hydrogen gas when zinc react with water. This expansion can result in extrusion of restoration beyond preparation margins and pulpal pain.

STRENGTH

- ▶ Exhibits high compressive strength and poor tensile strength and shear strength.
 - Compressive strength- 380-550Mpa
 - Tensile strength- 48-64Mpa
- ▶ Amalgam is brittle when placed in thin sections, b/c of its low edge strength. Hence amalgam should have adequate bulk.
- ▶ Increase in mercury, decrease the strength of amalgam.
- ▶ Spherical alloys are harder and stronger when compared to lathe cut alloys because they require less mercury during trituration, thus less of the weak matrix portion forms.

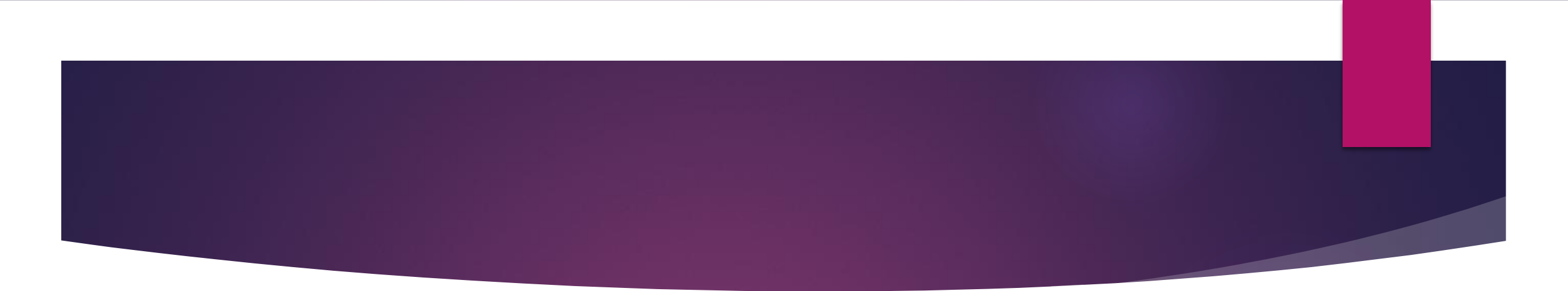


CREEP

- ▶ Creep is progressive permanent deformation of set amalgam under dynamic loading during function.
- ▶ Low copper amalgam-2.5% creep(>y2)
- ▶ High copper amalgam-0.2% creep(<y2)
- ▶ Clinically creep lead to protrusion of restoration making the amalgam more prone to fracture, overhangs

BIOCOMPATIBILITY

- ▶ It is related to the mercury vapour released during manipulation, placement and removal of amalgam.
- ▶ However the amount is minimal and the possibility of toxic reactions is less.
- ▶ Careful handling of mercury reduces the risk to dentist or dental assistant.



COEFFICIENT OF THERMAL EXPANSION

- ▶ It is three times more in amalgam than that of dentin. This large difference is responsible for microleakage.

THERMAL CONDUCTIVITY

- ▶ Because of good thermal conductivity, amalgam can transmit temperature changes readily to the pulp. Hence, its closeness to pulp should be avoided without adequate pulp protection.

Microleakage in Amalgam

- ▶ Microleakage occurs when there is 2 to 20 micron wide gap between the amalgam and tooth structure.
- ▶ The following factors are responsible for microleakage in amalgam:
 - 1 • Poor condensation techniques that cause marginal voids
 - 2 • Lack of corrosion by products which are necessary for sealing of margins
 - 3 • High coefficient of thermal expansion of amalgam
 - 4 • Use of single composition spherical alloys which show more, leakage than lathe-cut or admixed alloys.

2. PRINCIPLES OF TOOTH PREPARATION FOR AMALGAM RESTORATIONS

► A. Amalgam Restoration for Class I Tooth Preparations

The basic principles of tooth preparation are as follows

- ➔ **Initial tooth preparation:**
 - ◆ Outline form
 - ◆ Primary resistance form
 - ◆ Primary retention form
 - ◆ Convenience form.
- ➔ **Final tooth preparation:**
 - ◆ Management of remaining caries
 - ◆ Secondary resistance and retention form
 - ◆ Pulp protection, if required
 - ◆ Finishing of enamel margins
 - ◆ Final inspection of the preparation.

1. Outline Form

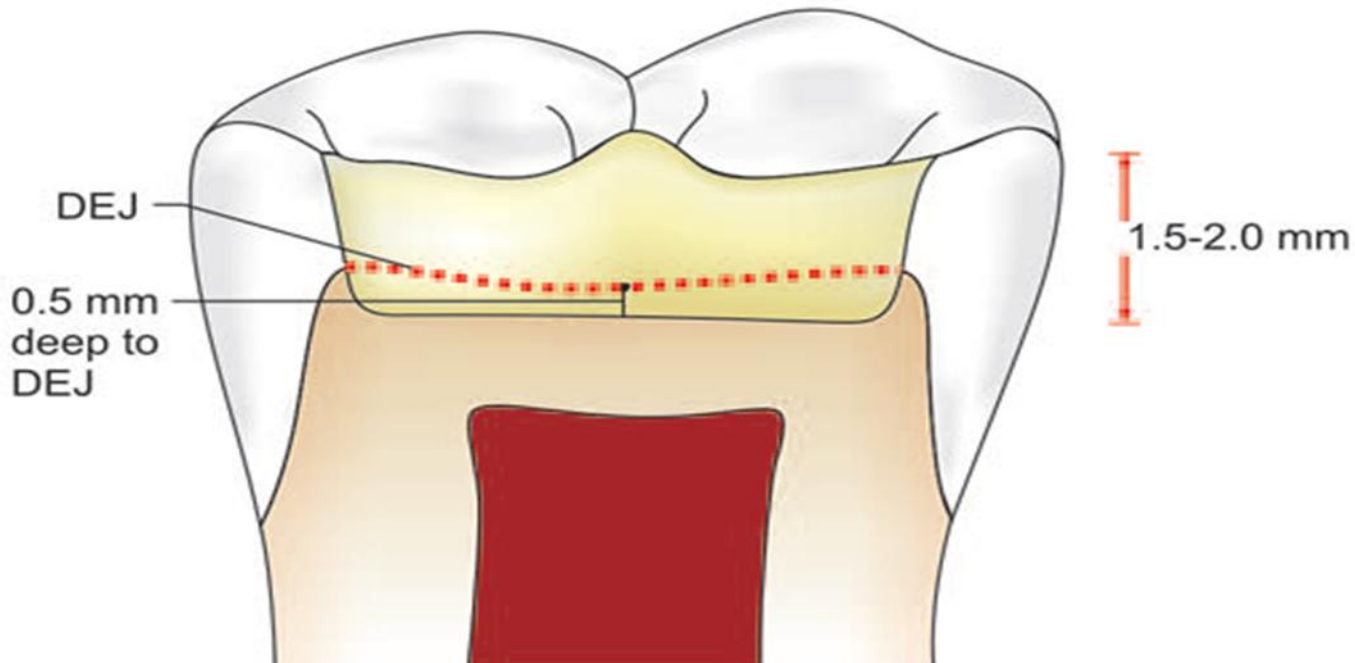
- ▶ The outline form means extending the preparation margins to the place they will occupy in the final preparation.

- ▶ NOTE
- ▶ Following facts, must be kept in mind while making outline form:
 - 1• Removal of all carious and defective pits and fissures to healthy tooth structure.
 - 2• Removal of all unsupported enamel rods
 - 3• To avoid ending preparation margins in high stress areas like cusp tip and crest of the ridges.
 - 4• Placing margins on sound tooth structure.

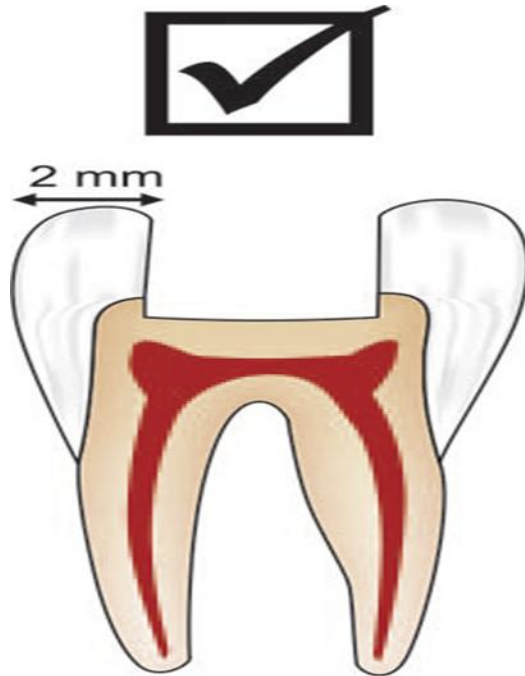
Steps

- ▶ 1 • Establish the external outline form to extend all margins into sound tooth tissue (With the help of no. 245 bur.)
- ▶ 2 • Bur should be kept parallel to long axis of the tooth to make a ditch in the carious portion of the tooth and it should be rotating when applied to the tooth and should not stop rotating until removed.
- ▶ 3. The external outline form should have smooth curves, straight lines and rounded angles. All unsupported and demineralized enamel should be removed.
- ▶ 4. The deep pit and fissure defects less than 0.5 mm apart should be included within the outline form.

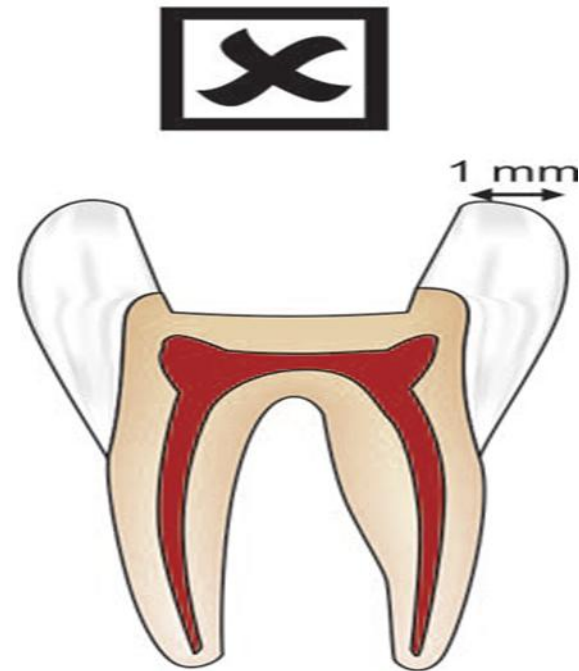
- 5• Maintain the initial depth of 1.5 mm, this is approximately one half the length of the cutting bur. This should be, at least 0.2 to 0.5 mm in dentin to provide adequate strength to resist fracture due to occlusal forces.



- ▶ 6• Extend the margin mesially and distally but do not involve marginal ridges. These walls should have dovetail shape to provide retention to the restoration.

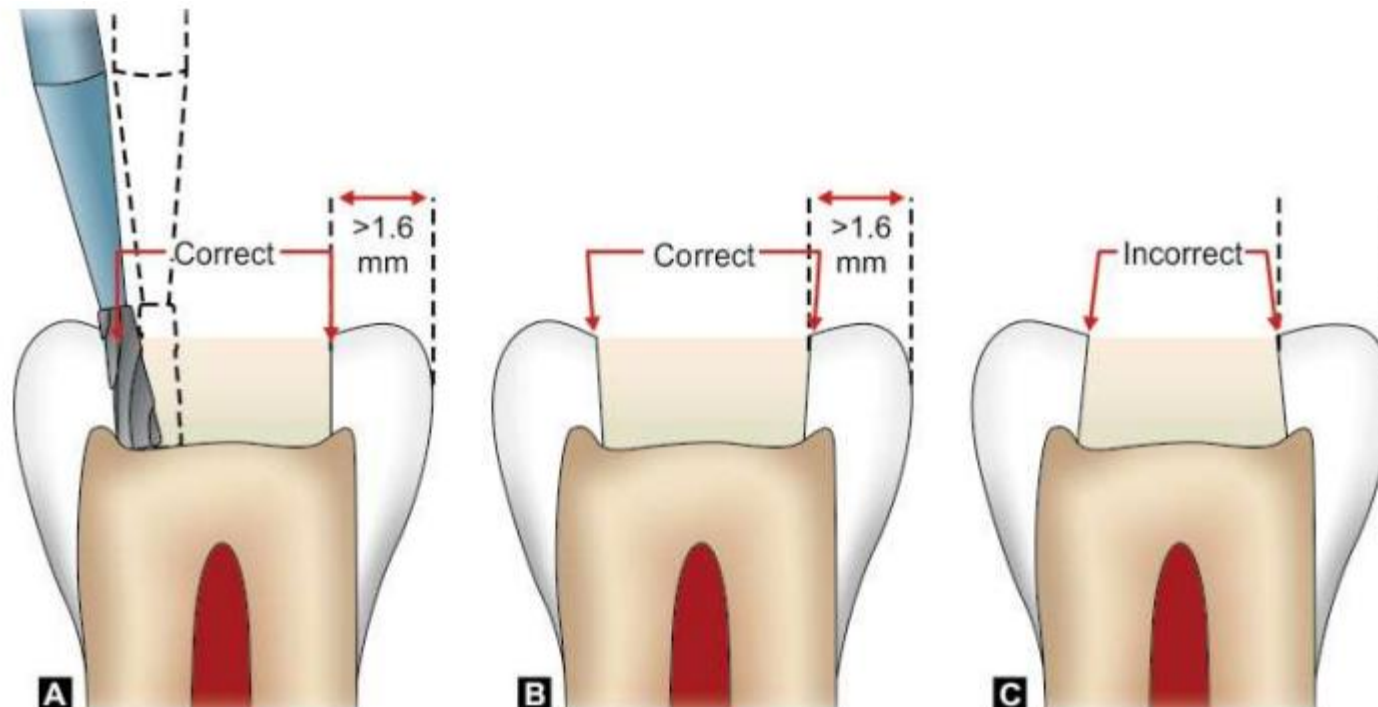


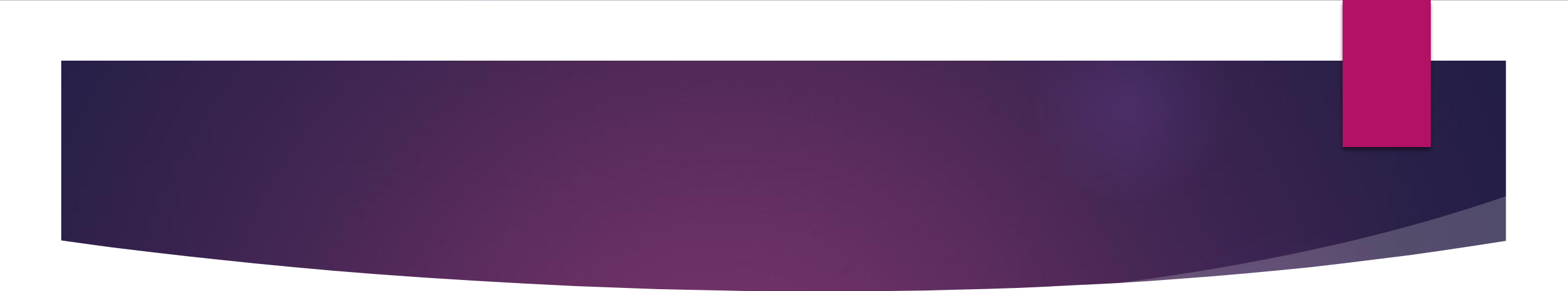
Adequate
marginal ridge



Avoid thinning
of marginal ridge

- 7. While working towards mesial and distal surface, orient the bur towards respective marginal ridge. This will result in slight divergence of mesial and distal walls which helps to provide dentinal support for marginal ridges.



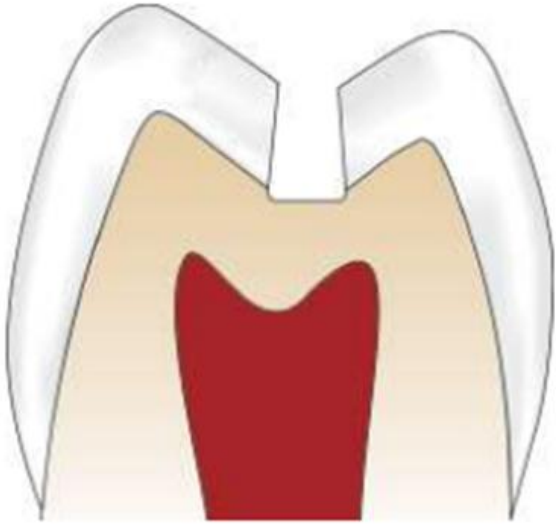
- 
- ▶ The isthmus width should be as narrow as possible, it should not be wider than the intercuspal distance. less than one-fourth ($1/4$) of the intercuspal distance
 - ▶ Enameloplasty is the careful removal of sharp and irregular enamel margins by 'rounding' or 'saucering' it and forming a easily cleansable area. The enameloplasty should not extend the outline form.
 - ▶ The use of enameloplasty should be done in ends of fissures whenever needed.

2. Primary Resistance Form

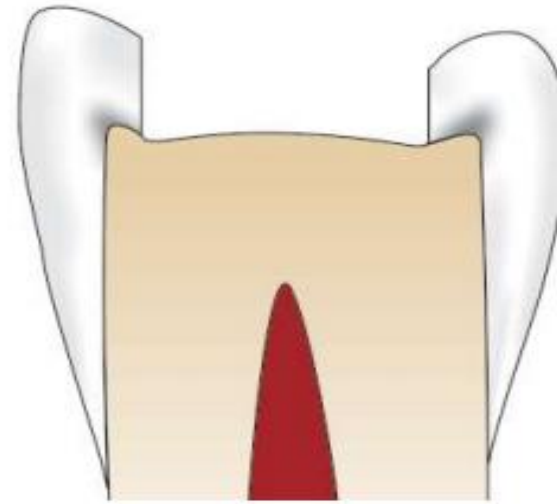
- ▶ Definition:
- ▶ Resistance form is that shape given to a preparation planned to afford such a seat for the restoration so as to best enable it to withstand the occlusal stresses.

Primary resistance form should have following features:

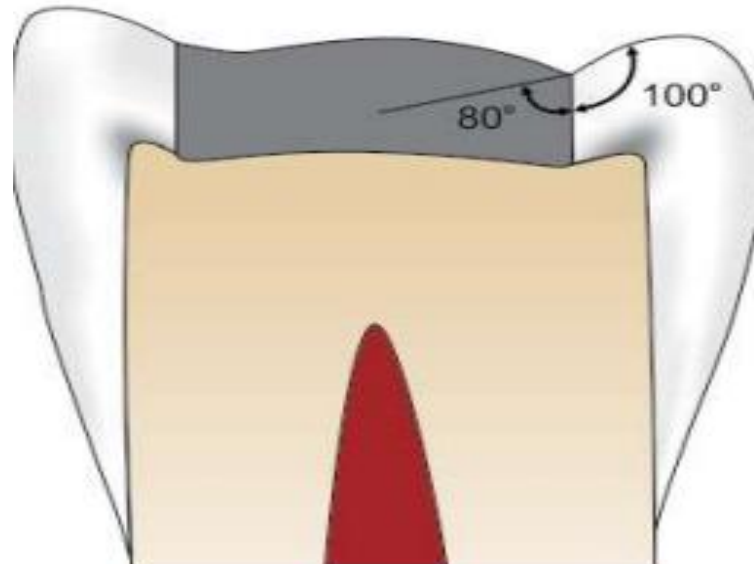
- ▶ Shape of the preparation should be like a box with flat floor. This helps the tooth to resist occlusal masticatory forces without any displacement.
- ▶ Though it should be flat, at the same time it should follow the contour of occlusal surface
- ▶ Provide the cavosurface angle of 90°



Box shaped preparation to provide resistance form



Though pulp floor is flat but it should follow the contour of occlusal surface

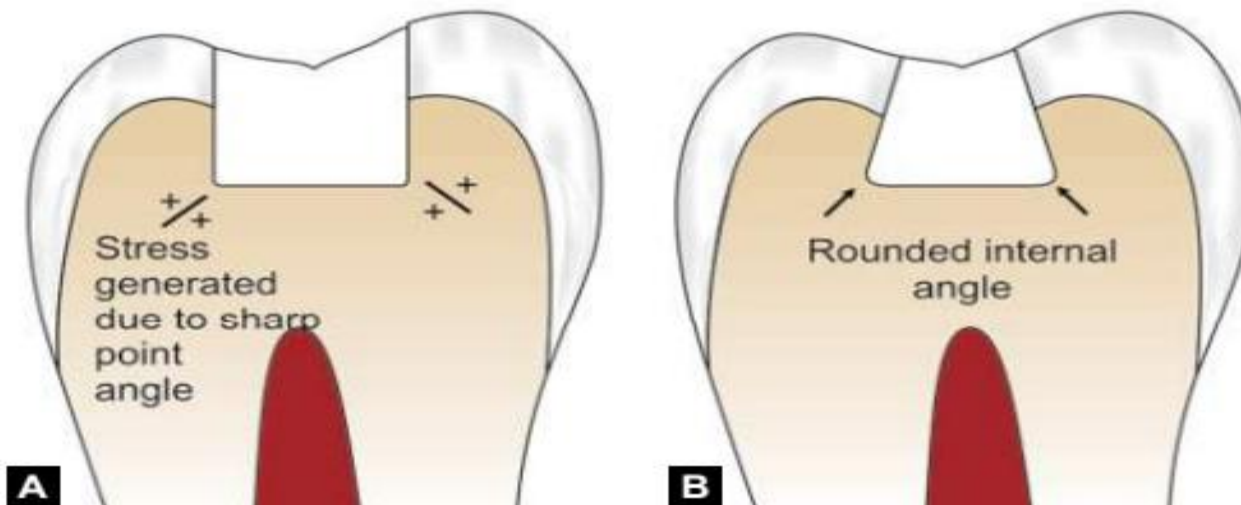


Amalgam-tooth interface should have butt joint

The minimum thickness varies by material:

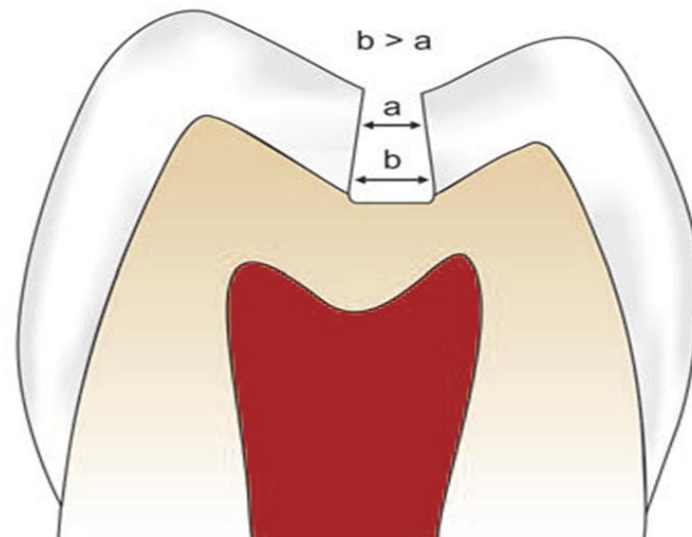
- **Amalgam:** Minimum of 1.5 mm.
- **Cast metal:** 1.0 to 2.0 mm.
- **Ceramics:** Minimum of 2.0 mm.
- **Composites and glass ionomers:** Usually require a thickness of more than 2.5 mm due to their occlusal wear potential

- ▶ Restrict the extension of external walls so as to have strong marginal ridge areas with sufficient dentin support
- ▶ Include all the weakened tooth structure
- ▶ Round off all the internal line and point angles
- ▶ Consider capping of cusp for preserving cuspal strength.

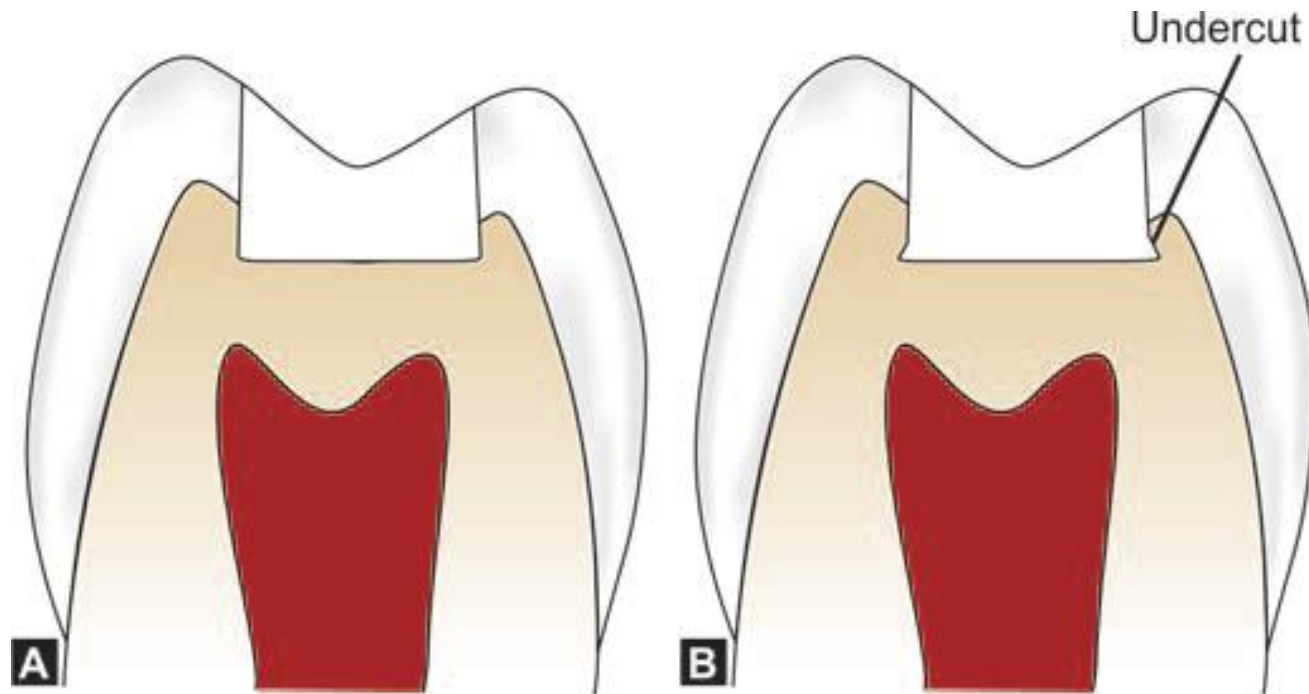


3.Primary Retention Form

- ▶ Definition
- ▶ Primary retention form prevents the restoration from being displaced.
- ▶ Retention can be Increased by the following:
- ▶ 1 • Occlusal convergence (about 2 to 5%) of buccal and lingual walls



- ▶ 2• Giving slight undercut in dentin near the pulpal wall.



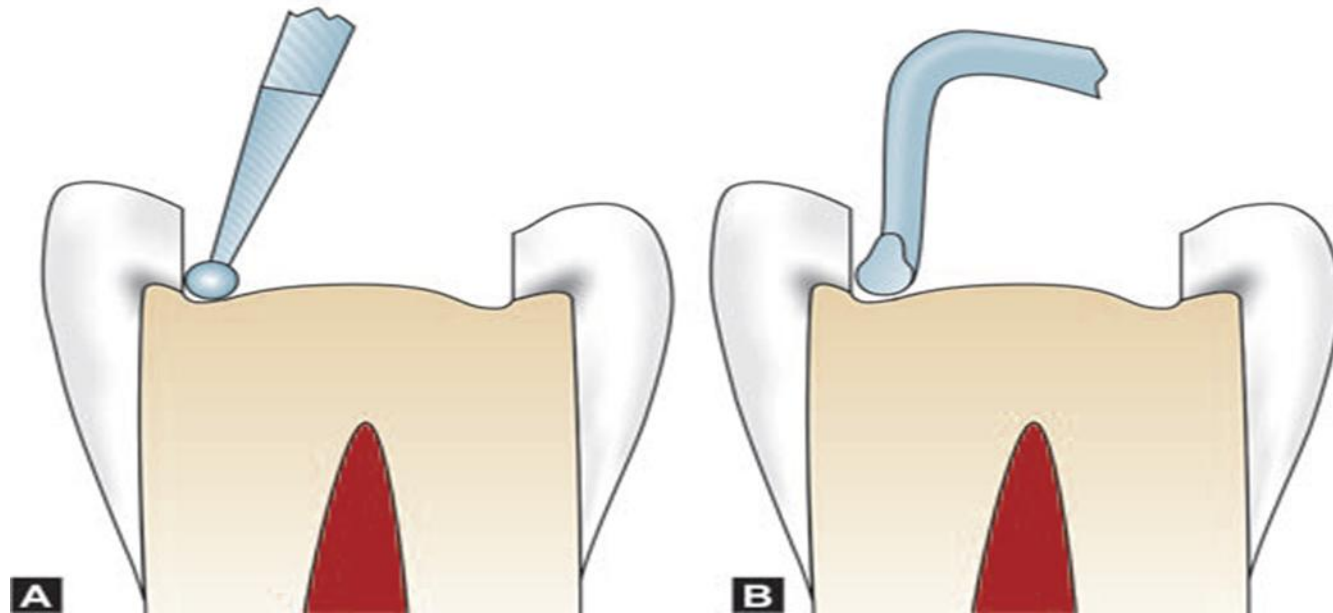
- 3• Conserving the marginal ridges.
- 4• Occlusal dovetail.

4. Convenience Form

- ▶ Definition :It is the form or shape that also permits access of condensing and carving instruments.
- ▶ The convenience form of the preparation facilitates and provides:
 - Sufficient visibility, accessibility and
 - Ease of operation in preparation and restoration of the tooth.

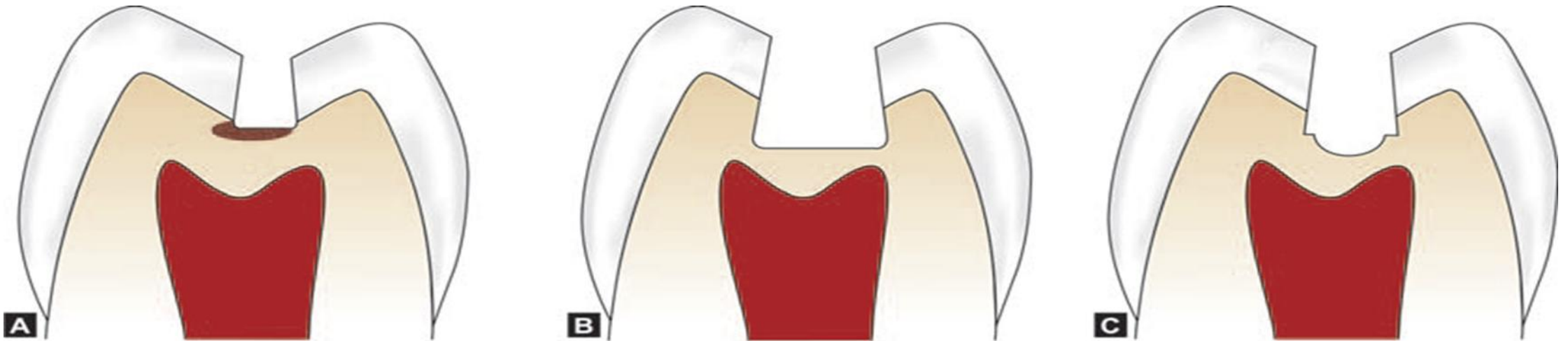
5. Removal of Remaining Carious Dentin

- ▶ In this, remaining caries, old restorative material and adjacent deep pits and fissures are also removed and involved in the preparation
- ▶ In the large preparations with soft caries, the removal of carious dentin is done with spoon excavator or slow speed round bur.



(A) Removal of remaining caries using round bur (B) Removal of caries using spoon excavator

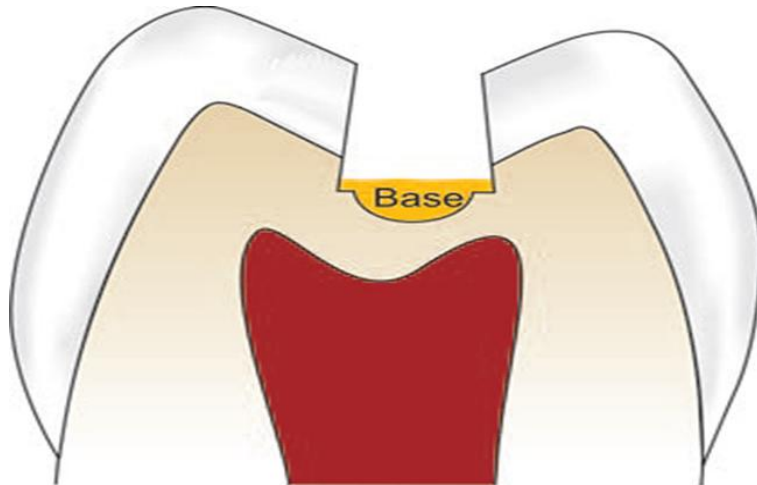
- ▶ In this, two-step pulpal floor is made, i.e. only portion of tooth which is affected by caries is removed, leaving the remaining, floor untouched.



(A) Caries present beyond tooth preparation (B) Overpreparation of tooth in an attempt to involve caries (C) Stepped pulpal floor to involve carious lesion

6. Protection of Pulp if Needed

- ▶ Use of pulp protective materials depends upon following factors:
- ▶ 1• A base is not needed in shallow preparations, as it compromises thickness and thus compressive strength of amalgam.
- ▶ 2• In a deep preparation, a base is placed on the deepest part in the thickness of 0.5 to 0.75 mm, so as to protect pulp.

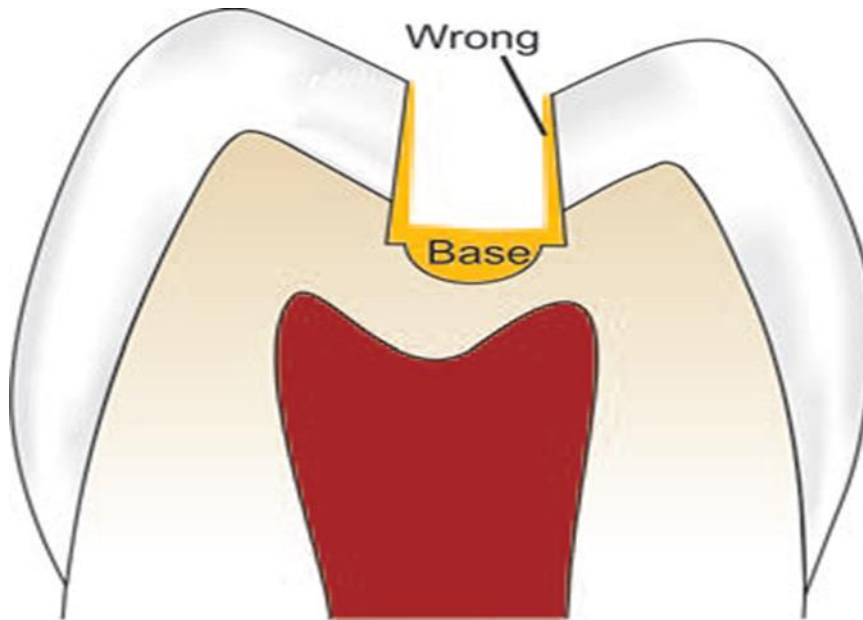


In deep preparations, base is applied on pulpal floor so as to protect pulp

NOTE

- ▶ The selection of bases for amalgam restorations are based on two factors:
 - ▶ 1• It should have sufficient strength to support the forces of amalgam condensation
 - ▶ 2• It should be able to strengthen the restoration under the masticatory stresses.
- ▶ The strength of the bases depends upon the size,thickness and mechanical properties of base along with preparation design, position and amount of occlusal load.

- Be sure that no trace of the base material remains on enamel walls of preparation , as this would eventually dissolve in the oral fluids leaving a gap between the restoration and the tooth resulting in microleakage and recurrent caries.



Do not place base on walls as it would ultimately dissolve in oral fluids, leaving a gap between tooth and restoration

7. Finishing of the Enamel Walls and Margins

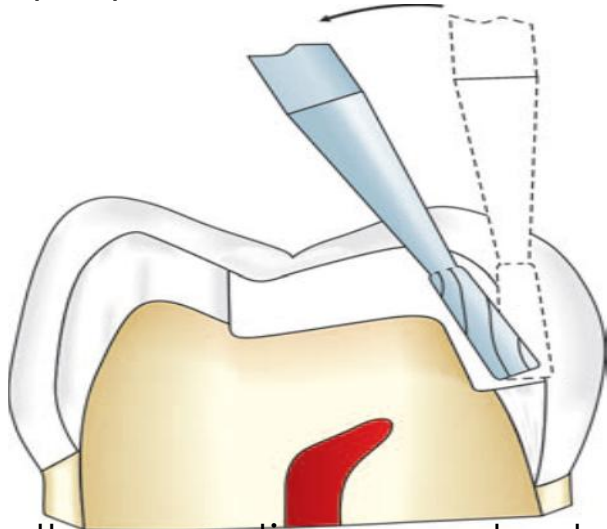
- ▶ Finishing of walls and margins is guided by the knowledge of dental histology. At this stage all unsupported enamel is removed.
- ▶ Cavosurface angle, i.e. angle between enamel wall and amalgam interface should be made 90° butt joint type. This provides bulk to restoration, which in turn provides maximum strength.

8. Final Cleaning and Inspection of the Preparation

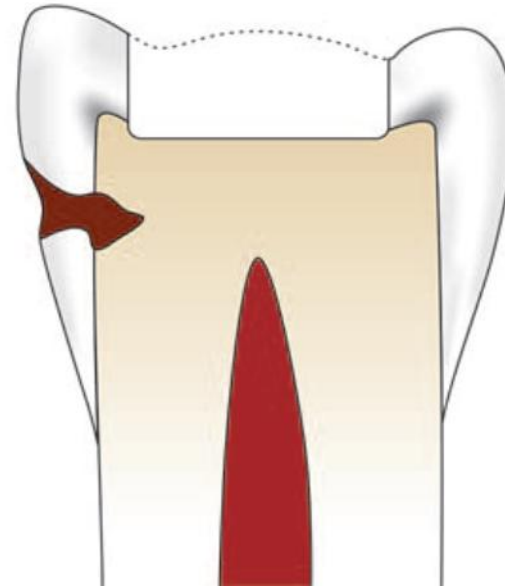
- ▶ The final stage of tooth preparation is to clean the preparation thoroughly with water and air spray.
- ▶ Then dry it with moist air and inspect it for final approval.

2. Class II Tooth Preparation for Amalgam Restoration

- ▶ Def: Class II restoration involves the proximal (mesial or distal) surfaces of premolars and molars.
- ▶ Two stages
- ▶ 1. Occlusal stage : Class II preparation is initiated same as Class I preparation i.e. entrance through occlusal surface.



Tooth preparation on occlusal surface with buccal or lingual extension



Make the occlusal box with the uniform depth of pulp floor

Outline form

- ▶ In the proximal area is primarily determined by:
- ▶ The faciolingual position of the contact area and
- ▶ The extent of carious lesion.

STEPS

▶ 1. Extend the preparation

- ▶ • Established pulpal depth, with the bur held parallel to the long axis of the tooth, to extend the preparation towards the contact area of the tooth, ending short by 0.8 mm of cutting through the marginal ridge.

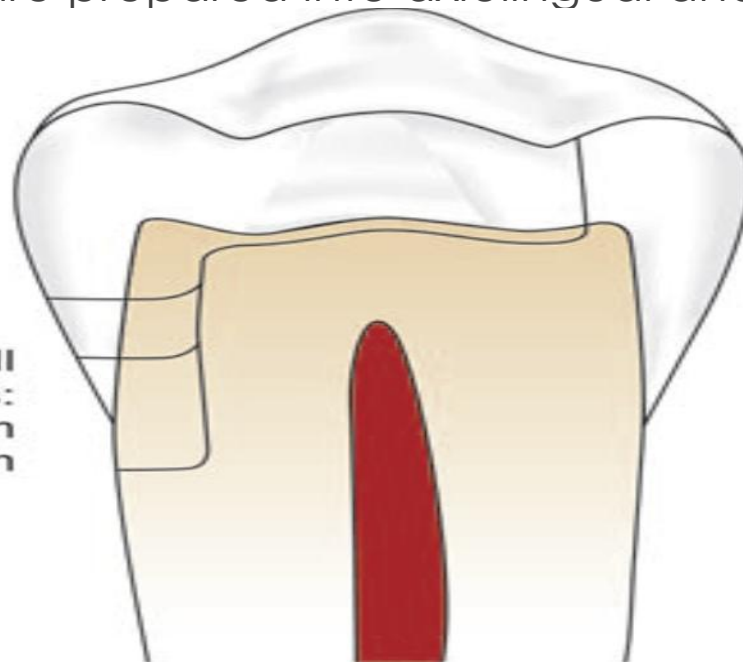


Extend the preparation ending short by 0.8 mm of cutting through marginal ridge

► 2. Proximal cutting

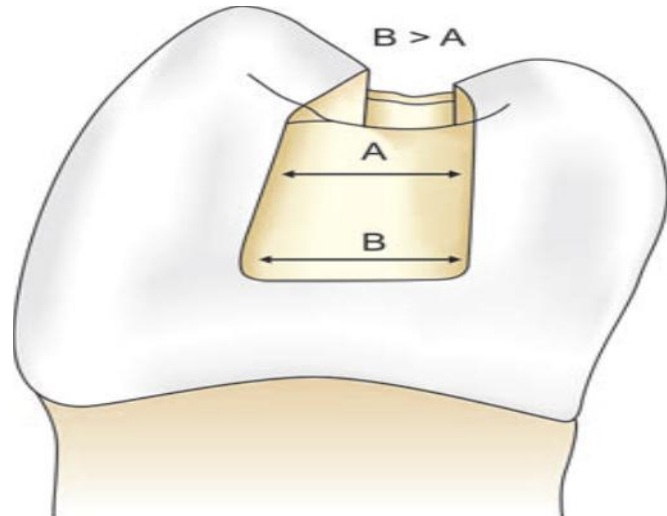
- The proximal cutting is sufficiently deep into the dentin (0.5-0.6 mm) so that retentive locks are prepared into axiolingual and axiofacial line angles.

Axial wall
dental depths:
crown, 0.5-0.6 mm
root, 0.75-0.8 mm



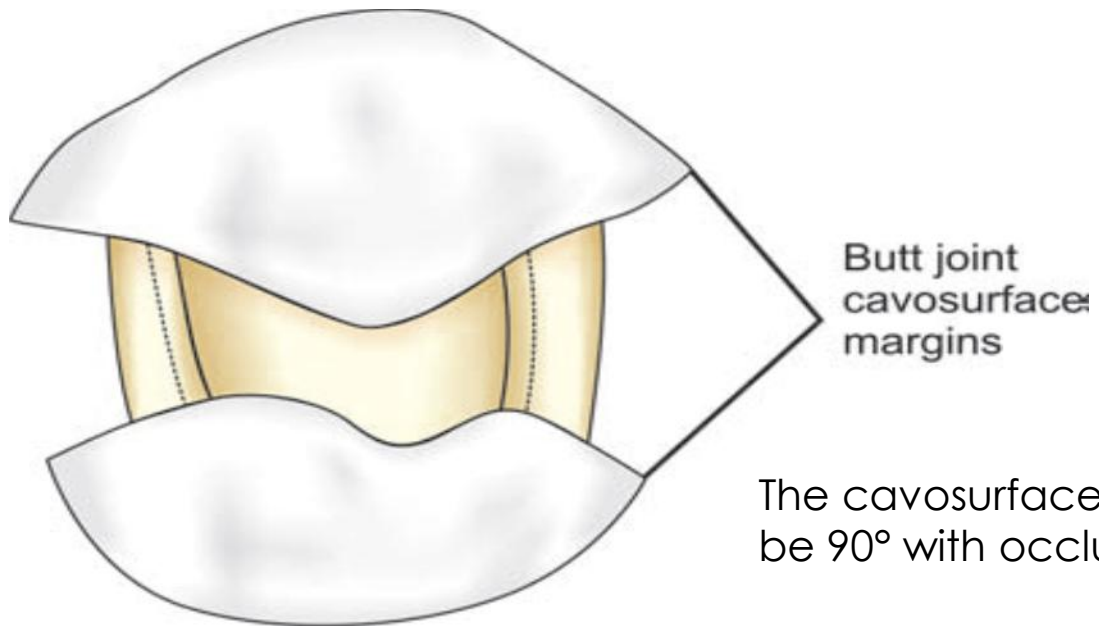
Proximal cutting should be
Sufficiently deep into dentin

- ▶ 3. Faciolingual
- ▶ Widen the preparation faciolingually to just clear the contact areas. The proximal cut is diverged gingivally.
- ▶ In other words, faciolingual dimension at the gingival surface is greater than the occlusal surface, this provides good retention and conservation of marginal ridge.

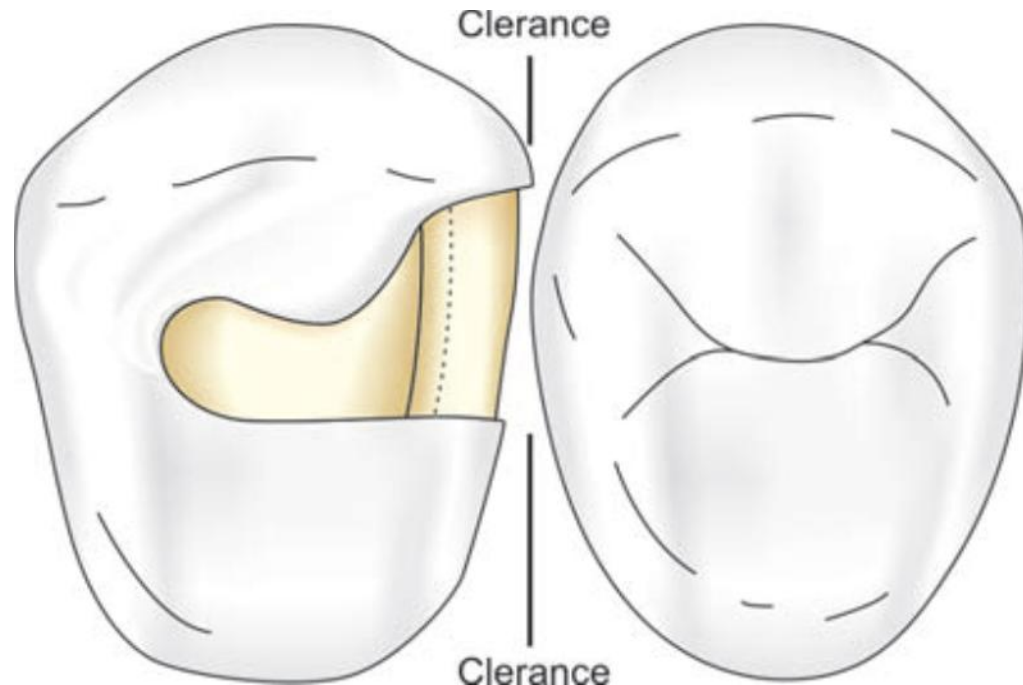


Faciolingual dimension of proximal box are more at gingival surface than at occlusal

- ▶ 4. Maintain proximal margins(PM)
- ▶ PM should have a cavosurface angle of 90° and when completed, the walls of the proximal box should converge occlusally.

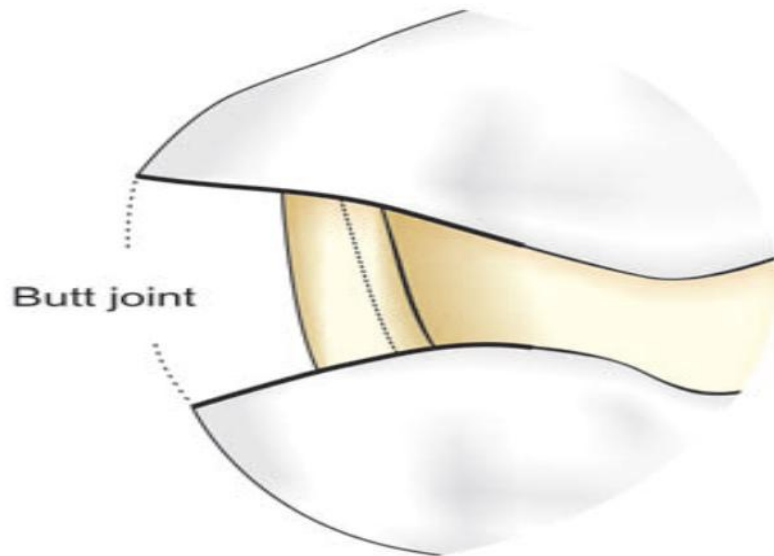


- ▶ 5. Proximal clearance
- ▶ Ideal clearance of facial and lingual margins of the proximal box should be 0.2 to 0.5 mm from the adjacent tooth.

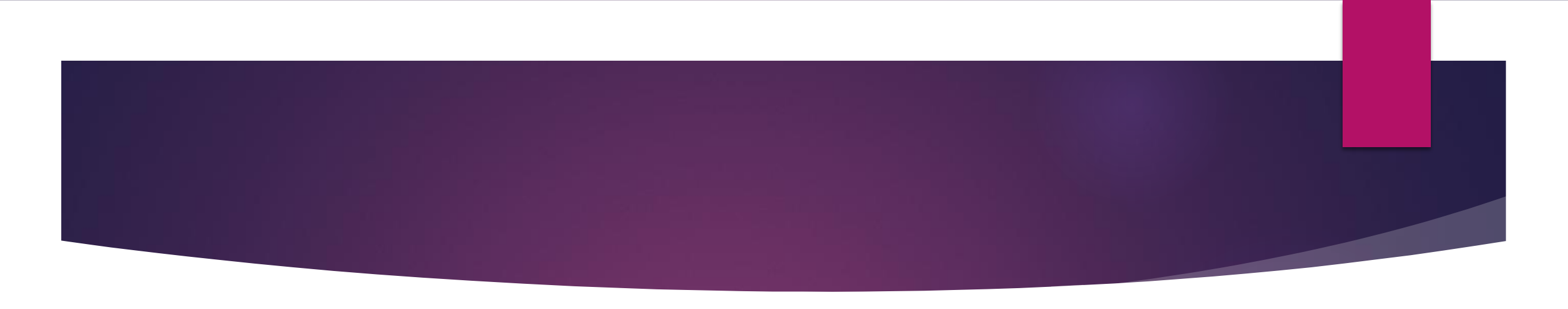


The ideal clearance of facial and lingual margins of proximal box should be 0.2 to 0.5 mm from adjacent tooth

- ▶ 6. Reverse curve
- ▶ is given to the proximal walls by curving them inwards towards the contact area. If excessive flare is given in these teeth, proximal walls will end past the axial angle of tooth through the cusps resulting in weakening of tooth structure and fracture of restoration.



Reverse curve is given to the proximal walls by curving them towards the contact area

- 
- ▶ **Reverse curve has following advantages{CPF}:**
 - ▶ Conserves the sound tooth structure.
 - ▶ Preserves the triangular ridge of the affected cusp.
 - ▶ Flare of the proximal wall leaves the tangent to that outer tooth surface at 90° angle, this causes further increase in resistance for both tooth and restoration.

2. Primary Resistance Form

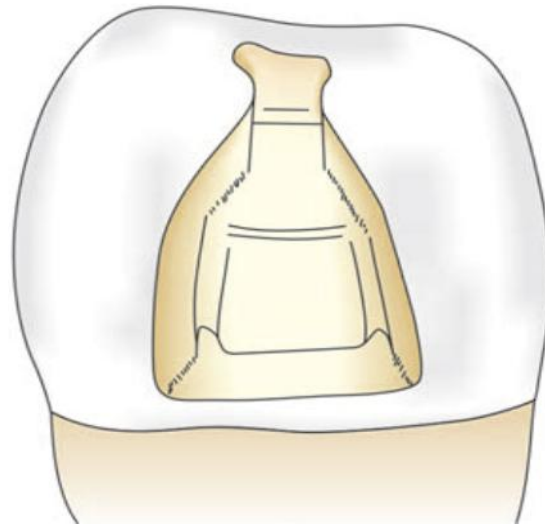
- ▶ This can be obtained by incorporating following features in the preparation:
- ▶ Shape of the preparation like a box with flat pulpal and gingival floor.
- ▶ Cavosurface angle of 90° .
- ▶ Include all the weakened tooth structure.
- ▶ Maintain minimal width of the preparation so as to preserve tooth structure.
- ▶ Round off all the internal line and point angles.
- ▶ Consider capping of cusp for preserving cuspal strength.

3. Primary Retention Form

- ▶ Primary retention form prevents the restoration from being displaced.

Retention can be increased by the following:

- ▶ Occlusal convergence (about 2 to 5%) of buccal and lingual walls.
- ▶ Occlusal dovetail.



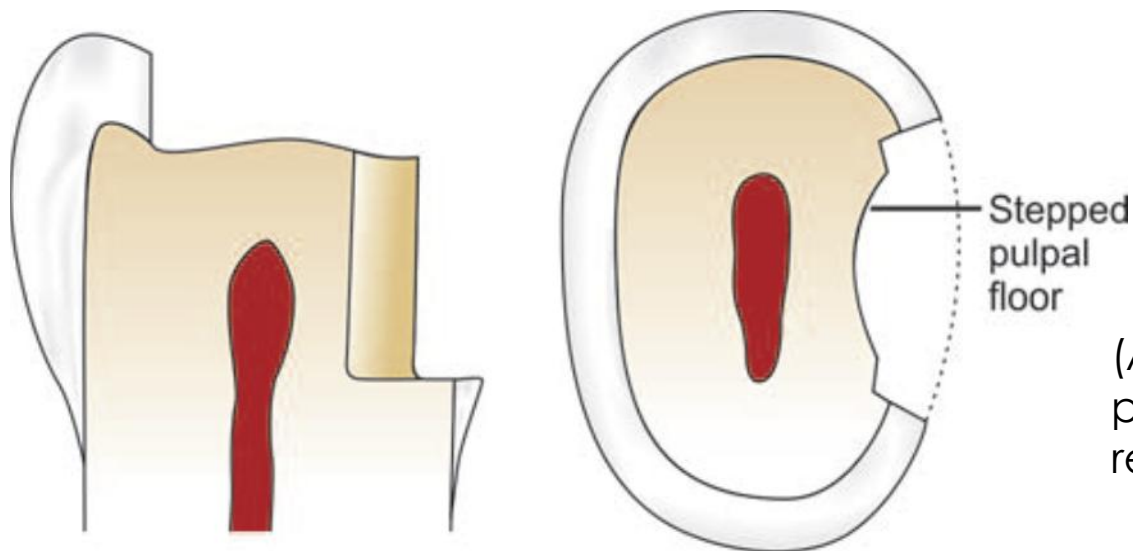
Occlusal convergence of buccal and lingual walls provide retention to amalgam restoration

Final Tooth Preparation

- ▶ During final preparation of tooth:
- ▶ Clean it with air/ water spray or with cotton pellet and
- ▶ Inspect it for detection and removal of debris and examine for correction of all cavosurface angles and margins.
- ▶ Remove remaining caries, old restorative material and adjacent deep pit and fissure involved in the preparation as done in class I preparation.

NOTE

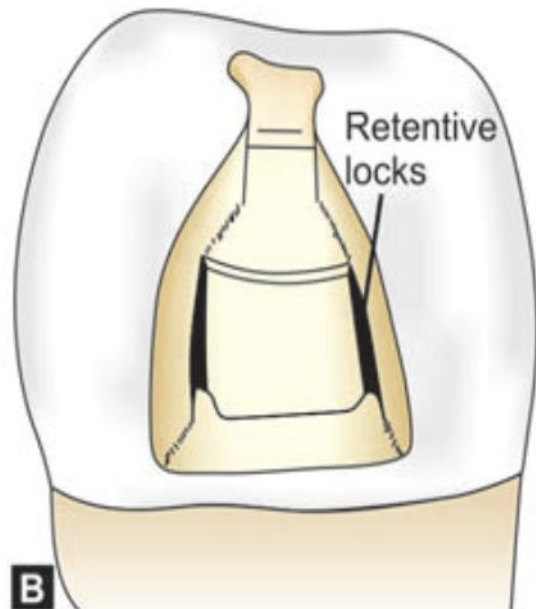
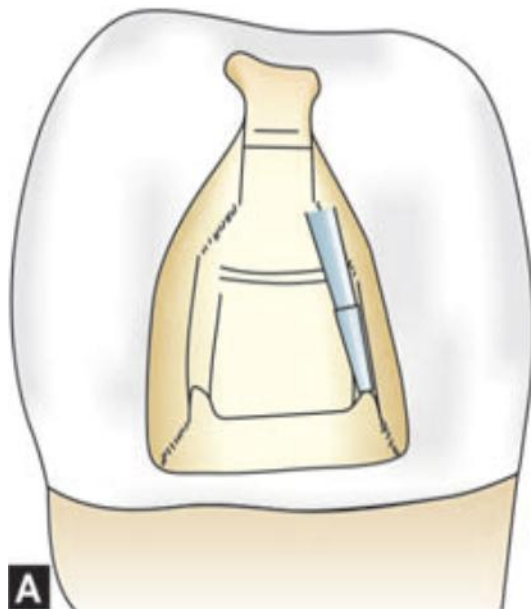
- ▶ In the large preparations with soft caries, the removal of carious dentin is done with spoon excavator or slow speed round bur.
- ▶ In this, two step pulpal floor is made, i.e. only portion of tooth which is affected by caries is removed, leaving the remaining preparation floor untouched.



(A) Caries in axial wall extending towards pulp (B) Removal of carious dentin only, resulting in stepped pulpal floor

Secondary retention and resistance form

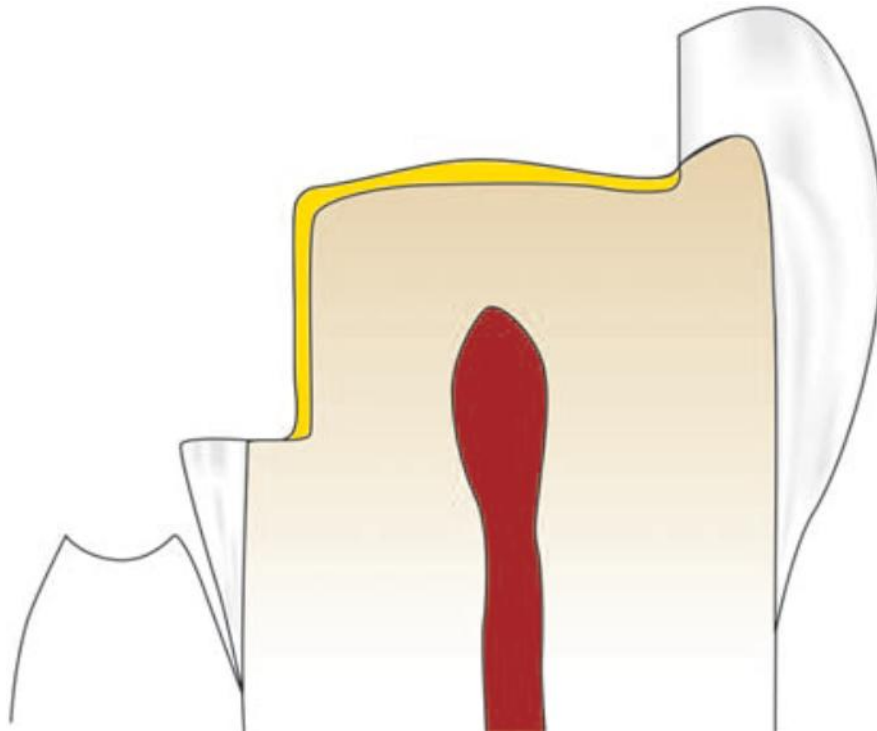
- ▶ Placing retention grooves and locks in the proximal box.
- ▶ 'Slots' and 'Pot holes' in the gingival floor may be given to provide additional retention.



Retention locks in the proximal box (A)
Preparation of locks using fissure bur (B)
Completed locks in proximal box

Pulp protection

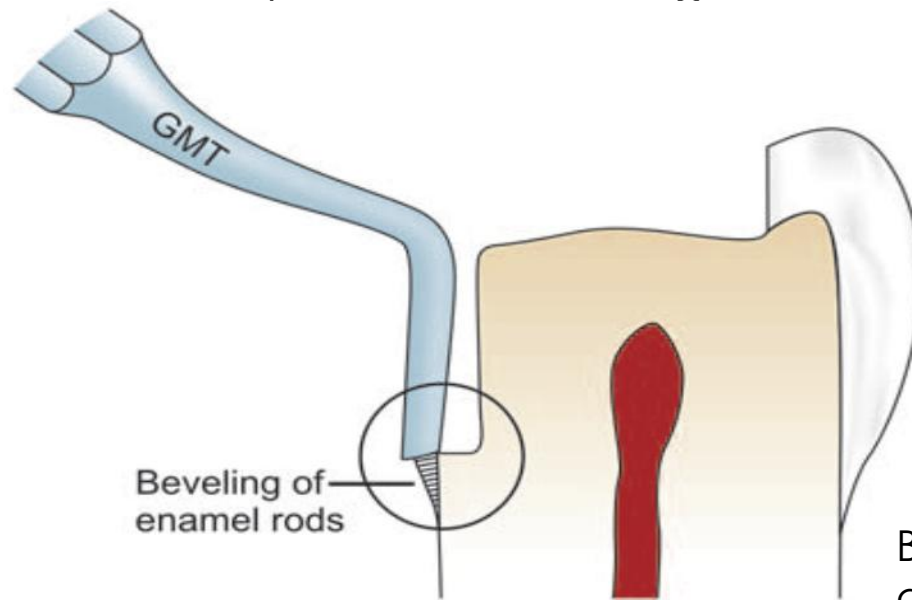
- • Use pulp protective materials whenever needed as in class I restoration.



Base is applied on pulpal and axial wall in class II preparation

Finishing

- ▶ Finally finishing of walls and margins is done by removing all unsupported enamel.
- ▶ Beveling of enamel portion of gingival wall is done with the help of gingival marginal trimmer. This helps to have full length enamel rods at the gingival margin.



Beveling of enamel portion of the gingival wall is done with the help of GMT

3. STEPS OF AMALGAM RESTORATION

Steps of amalgam restoration

- ➔ Selection of amalgam alloy
- ➔ Mercury—alloy ratio
- ➔ Trituration
- ➔ Mulling
- ➔ Application of matrix bond
- ➔ Insertion of amalgam
- ➔ Condensation
- ➔ Burnishing
- ➔ Carving
- ➔ Finishing and polishing.

1. Selection of Amalgam Alloy

- ▶ Following factors are considered while selecting an alloy for restoration:
- ▶ **Type of alloy:**
 - High copper or low copper alloys
 - Zinc free or zinc containing alloys
 - Size and shape of the particles.
- ▶ **Occlusal stress**
 - If restoration undergoes high occlusal stresses, choose amalgam with high resistance to marginal fracture

► **Patients psychology**

- Pt' with psychological problems or other diseases, requiring early disposal, indicate the use of fast setting alloy

► **Type of cavity preparation**

- In wider and broader preparations, alloy with low creep values is preferred.

► **Moisture control**

- If it is difficult to control moisture it is preferred to use zinc-free alloy to avoid delayed expansion.

2. Mercury : Alloy Ratio

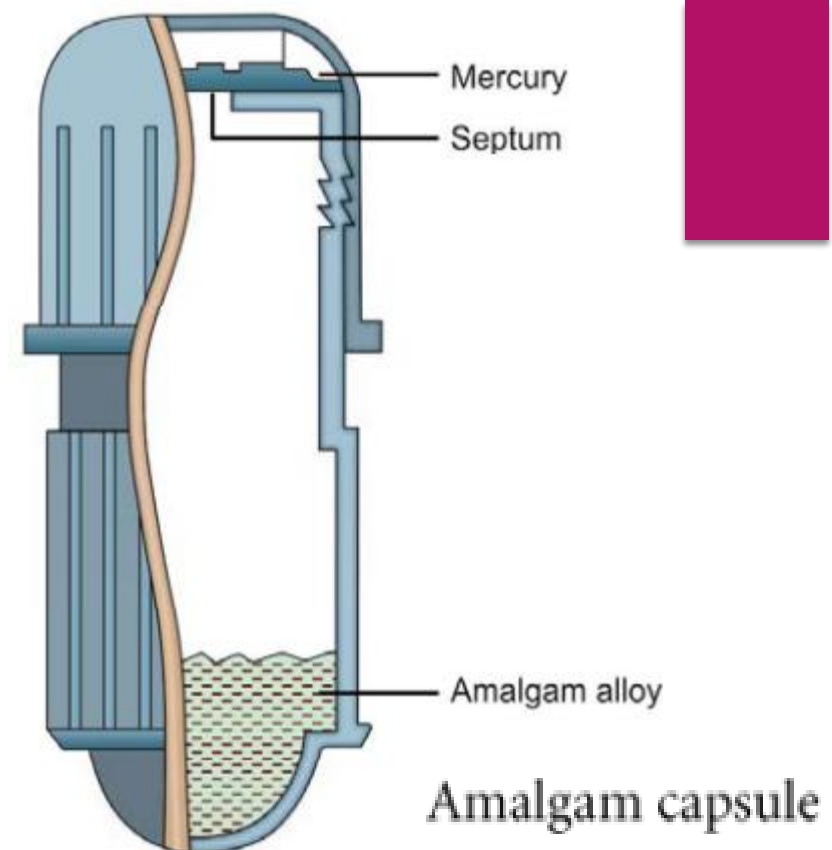
- ▶ For success of the restoration, mercury ratio should be specific and accurate are according to type of alloy used.
- ▶ Mercury is basically required to wet the alloy particles before they can react.
- ▶ Generally, it is 5:8 or 5:7, if mercury content is more than required amount, resultant mix will be weaker, but if it is less, it might not sufficiently wet the alloy particles.
- ▶ Lathecut amalgam alloys require more (45%) of mercury to wet than the spherical alloys (40%).

3.Trituration

- ▶ The purpose of trituration is to remove oxide layers from the alloy particles so as to coat each alloy particle with mercury, resulting in a homogeneous mass for condensation.
- ▶ Trituration can be done by hand or mechanical means. Mechanical method is done with the help of automatic amalgamator and hand method of trituration is done with the help of mortar and pestle



Amalgamator



Amalgam capsule



Properly mixed amalgam should be shiny, cohesive
homogeneous mass





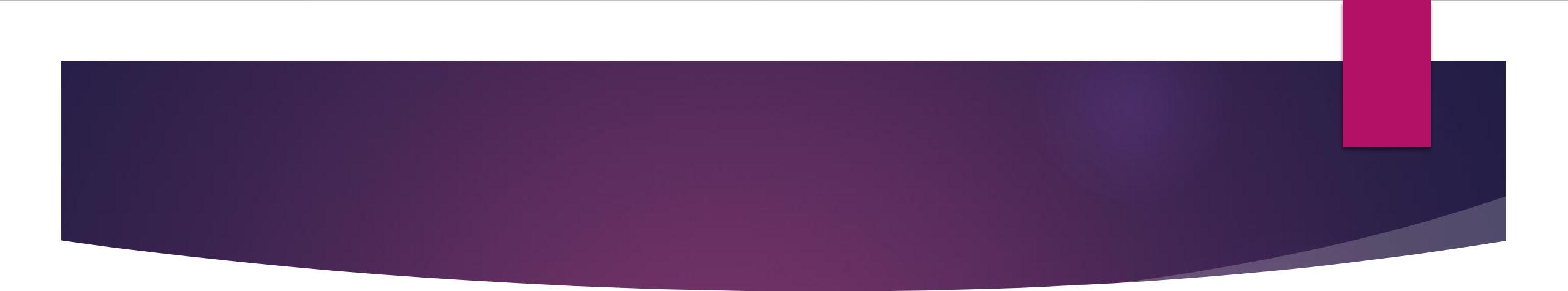
Undertritured amalgam mix appears dry and crumbled



Mortar pestle used for hand trituration

Objectives of trituration

- ➔ Achieve a workable mass of amalgam within minimum time
- ➔ Increase direct contact between the particle and mercury by removing oxides from powder
- ➔ Reduce particle size of powder so that fast and more complete amalgamation can be done
- ➔ Help in dissolving the particles of powder in mercury
- ➔ Reduce the amount of gamma-1 and gamma-2.

- 
- ▶ Time for which the trituration is carried out, speed and force applied for trituration, affect the quality of trituration.
 - ▶ Signs of a good mix amalgam is shiny, homogeneous mass that adheres together. Undertrituration results in a crumbly mix that is very weak. It causes decrease in the tensile and compressive strength values and increase in creep.
 - ▶ Overtrituration results in a mix that is warm and has a shiny surface. This overtritured mix sticks to the capsule which is difficult to remove. There occurs fast setting because the amalgam mass becomes heated.
 - ▶ Overtrituration causes increase in contraction, creep, tensile and compressive strength values for lathe-cut alloys, decrease in tensile and compressive strengths for spherical alloys.

Test for trituration

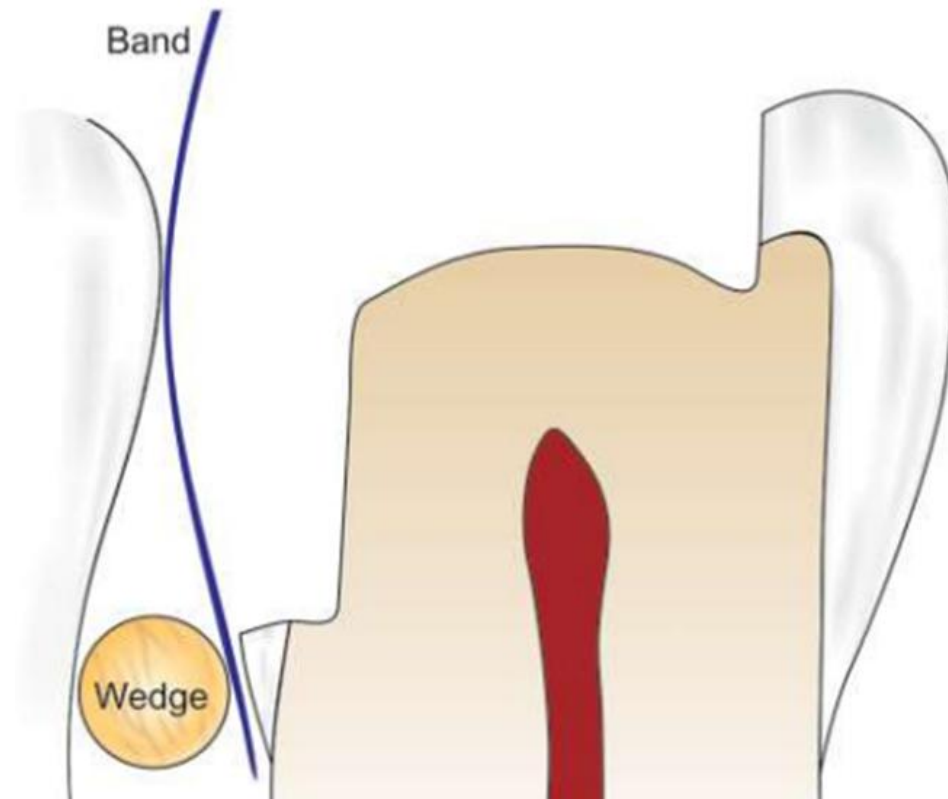
- ➔ Normal trituration
 - ◆ Good shiny mix
 - ◆ Homogeneous mass, adheres together.
- ➔ Over trituration
 - ◆ Mix is 'warm'
 - ◆ Difficult to remove from capsule
 - ◆ Shiny wet and soft.
- ➔ Under trituration
 - ◆ Dry mix
 - ◆ Crumbled mix that is very weak.

4. Mulling

- ▶ It is continuation of the trituration to create all alloy particles are properly coated with mercury. This can be done manually or mechanically. By hand, it can be done by squeezing the freshly mixed amalgam collected in the chamois skin.
- ▶ Mulling should not be done by bare hands as it can be contaminated by moisture. Mechanical mulling is done in the amalgamator by triturating it for one to two seconds.

5. Application of Matrix Band

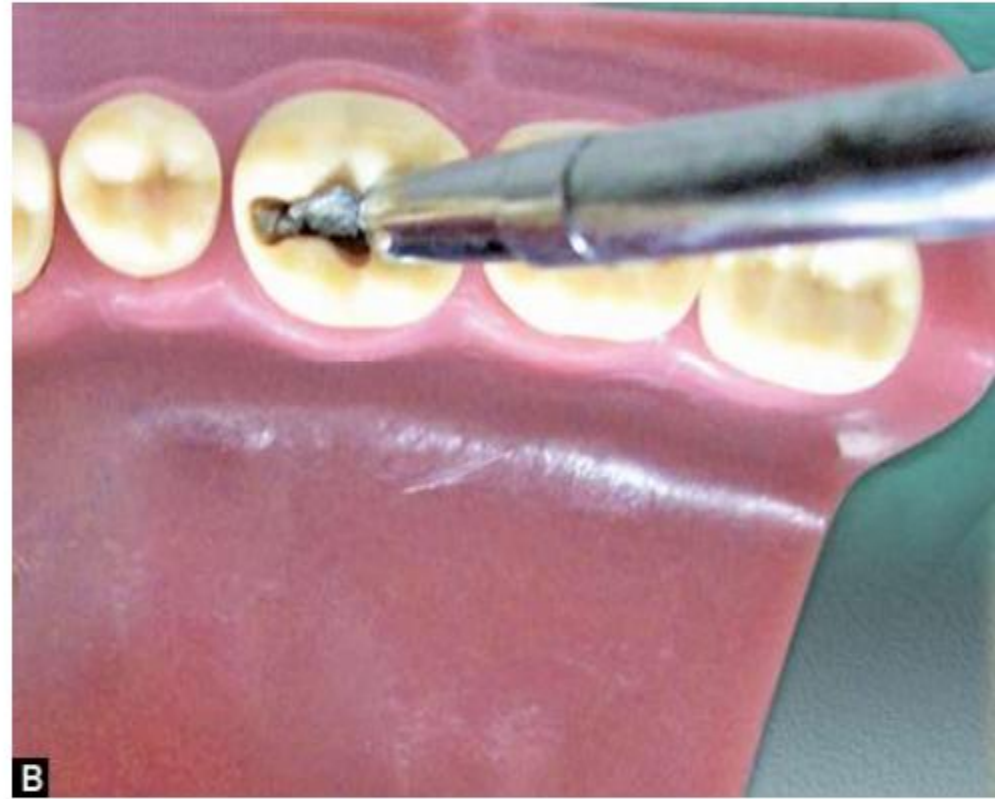
- ▶ A matrix is helpful to prevent “land-sliding” during condensation and to ensure marginal adaptation and strength of the restoration.
- ▶ The Tofflemire matrix retainer is used to secure a matrix band to the tooth. Place the matrix band in the matrix retainer.



Placement of band and wedge

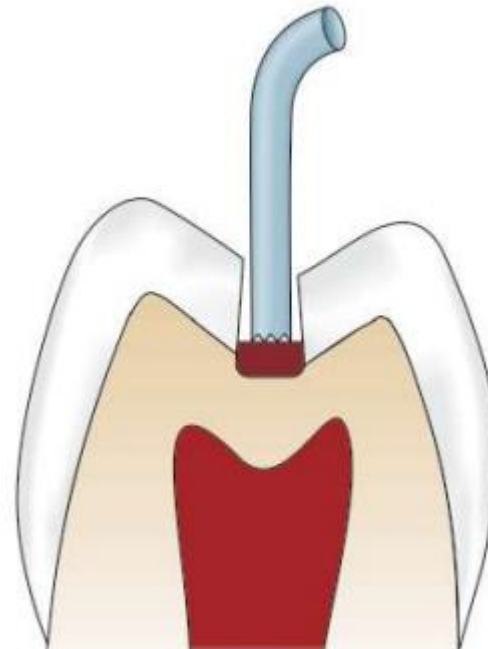
6. Insertion of Amalgam

- ▶ A small amount of amalgam alloy with the help of amalgam carrier and transfer it to the preparation.
- ▶ Proximal box should be filled before the occlusal part of the preparation.
- ▶ Place the first increment of amalgam in the deepest proximal part of the preparation and condense it with flat surface of condenser.
- ▶ Apply firm pressure on the amalgam mass for adequate condensation. After it, add next increment and again condense it.
- ▶ When the level of amalgam reaches the preparation margins, continue the packing of preparation to allow an excess to build up for better finishing.



(A) Class I preparation in 46, 47; (B) Insertion of amalgam into the preparation

- ▶ 7. Condensation
- ▶ Various shapes (triangular, round, elliptical, trapezoidal and rectangular) and sizes of condensers are used for amalgam condensation.
- ▶ Working end of a condenser is usually serrated.



Rules of condensation

- Start condensation within three minutes of trituration
- Condense continuously
- Condense laterally as well as apically
- Apply adequate force for condensation
- Have a constant supply of amalgam.

Objectives of condensation

- Brings excess mercury on the surface of restoration
- Reduces the number and size of voids in the restoration
- Prepares the surface of restoration for carving
- To adapt amalgam to the preparation walls and floors.

Condensation Depends upon Following Factors:

- ▶ Plasticity of the mass
- ▶ Size of the amalgam increment:
 - A larger mass results in incomplete condensation
- ▶ Condenser size:
 - The smaller the condenser working end, the greater the force
- ▶ Direction of force
- ▶ Amount of force
- ▶ Type of alloy:
 - Use larger condensers when condensing spherical alloys because smaller condenser will displace the spherical particles rather than condensing them.

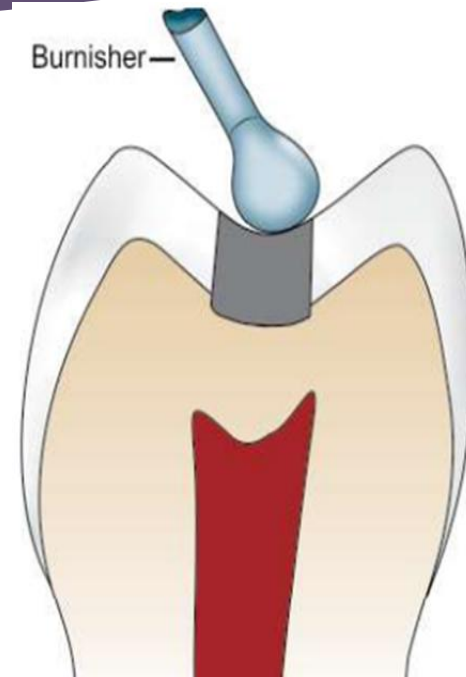
8. Burnishing : It is the process of rubbing, generally done to make the surface shiny.

► **Precarve burnishing**

- It is done after condensation. Amalgam is overfilled and burnished immediately with heavy strokes so as to improve marginal adaptability of the restoration and remove excess mercury from over packed amalgam .

► **Postcarve burnishing**

- It is done after completion of carving with the help of small sized burnishers using light strokes.



Advantages of precarve burnishing

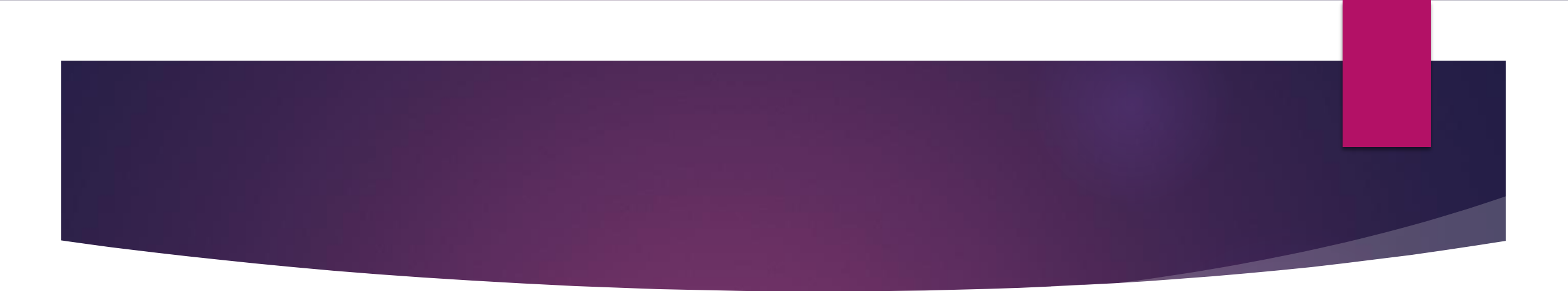
- Improves the marginal integrity of restoration
- Shapes the restoration according to contours and curvatures of the tooth
- Helps in reducing the mercuric content of amalgam.

Advantages of postcarve burnishing

- Reduces number of voids on surface of restoration
- Produces denser amalgam at margins
- Improves marginal seal
- Increases surface hardness
- Decreases rate of corrosion.

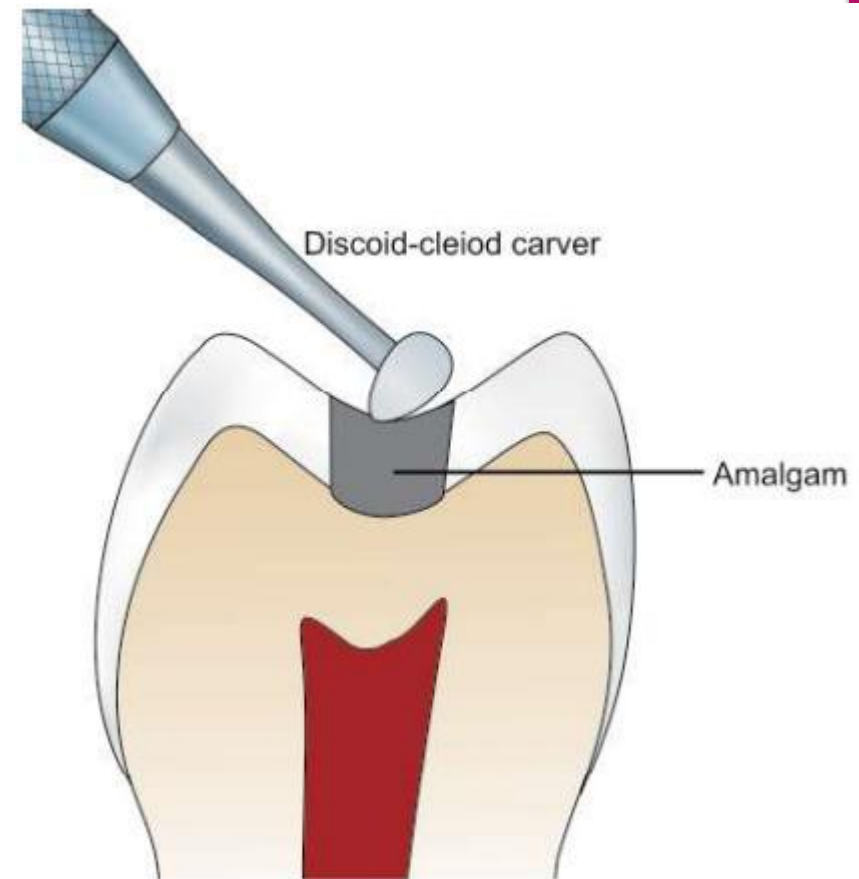
9. Carving

- ▶ Amalgam should not be carved until it is sufficiently firm. For adequate carving, it is preferable to over pack the preparation and then carve it to the margins.
- ▶ Carving causes removal of mercury rich surface layer. For proper carving, occlusal anatomy should be kept low to preserve bulk of the alloy at the margins.
- ▶ The carving instruments should have discoid and cleoid blade design. Larger instrument is used first, followed by smaller instruments.

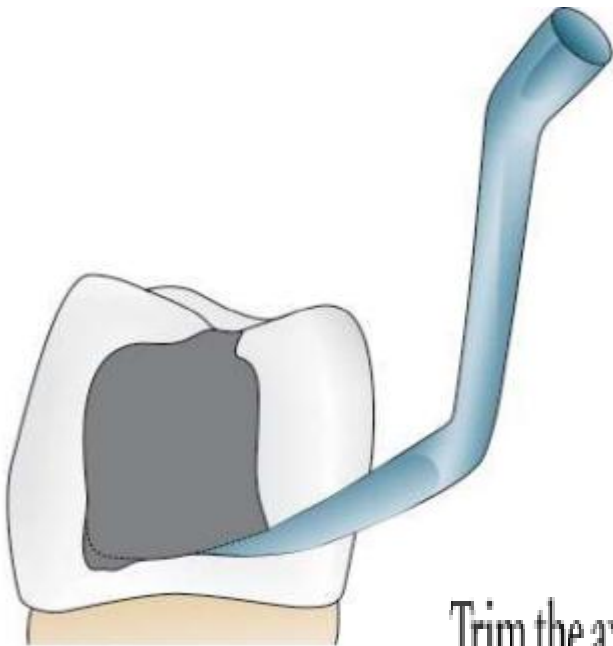
- 
- ▶ In proximal tooth preparation, carving of the cervical margins should begin following the removal of matrix band. Loosen the band and then wedge before carving because axial and cervical margins become accessible after removal of band.
 - ▶ Trim the axial margins towards the gingiva in downward direction with a sharp carver. Do not over carve amalgam as it can lead to acute angles and stress concentration within the amalgam which can fracture the restoration



Carving amalgam restoration



Use of discoloid-cleoid carver for carving occlusal surface



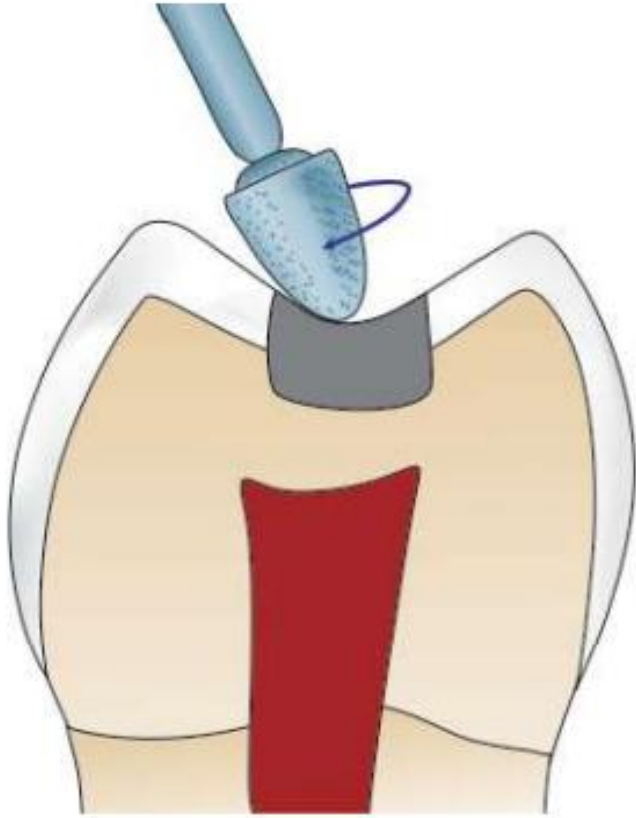
Trim the axial margins towards gingiva with sharp carvers

10. *Finishing and Polishing*

- ▶ Finishing and polishing of the restorations should be done at least 24 hours after the placement of the amalgam.
- ▶ Premature finishing and polishing will interfere with the crystalline structure of the hardening amalgam.

Advantages of finishing and polishing

- ➔ Improves marginal adaptation of restoration by removing flash
- ➔ Reduces tarnish and corrosion
- ➔ Polished surface is plaque resistant
- ➔ Polished surface is smoother and easier to clean
- ➔ Prevention of recurrent decay
- ➔ Prevention of amalgam deterioration
- ➔ Maintenance of periodontal health
- ➔ Prevention of occlusal problems.



Final polishing of amalgam restoration using
abrasive stones



Completed amalgam restoration

4. FAILURES OF AMALGAM RESTORATIONS

- Most common failures associated with dental amalgam restoration can be classified as;

Most common failures associated with dental amalgam restoration

→ *At microscopic level:*

- ♦ Pain after amalgam restoration
- ♦ Periodontal tissue injury due to proximal overhangs
- ♦ Pulpal involvement
- ♦ Tarnish and corrosion
- ♦ Internal stresses due to excessive masticatory forces.

→ *At macroscopic level:*

- ♦ Bulk fracture of restoration
- ♦ Tooth fracture
- ♦ Marginal fracture of amalgam
- ♦ Secondary or recurrent caries commonly takes place due to marginal leakage.
- ♦ Dimensional changes especially in zinc containing amalgam
- ♦ Discoloration of restoration
- ♦ Discoloration of tooth.

Reasons for Failure of Amalgam Restorations

The reasons for failure of amalgam restorations can be divided under following headings:

- ➡ Poor case selection
- ➡ Defective tooth preparation
- ➡ Defective amalgam manipulation
- ➡ Defective matrix adaptation
- ➡ Postrestorative failures.

1. Poor Case Selection

- ▶ For long-term success of the amalgam restorations, amalgam requires sufficient sound tooth structure to provide sufficient resistance and retention form.
- ▶ Selecting teeth with extensive caries, abnormal habits like bruxism and heavy masticatory forces can result in restoration failure. Thus amalgam should be placed in small to moderate size carious lesions.

2. Defective Tooth Preparation

- ▶ This is one of the major causes of failure of amalgam restorations. The following defects usually occur during tooth preparation.
- ▶ **Inadequate occlusal extension:**
 - Insufficient extension to include adjacent deep pits and fissures increases chances for secondary/recurrent caries

► **Under extension of the proximal box:**

- To prevent occurrence of secondary caries, walls of the proximal box of class II preparation must be extended to self cleansing areas.
- If the proximal margins of the filling are not adequately extended into the embrasures, they are not open to cleaning by mastication and brushing resulting in secondary caries.

► **Over extension of the proximal box:**

- On the other hand, over extension into the embrasure areas makes the preparation walls weak specially in lower bicuspid and on distal sides of maxillary and mandibular first molars.
- **Depth of preparation:** Minimum depth of preparation should be 1.5 to 2 mm so as to provide bulk which can prevent its fracture under masticatory load.

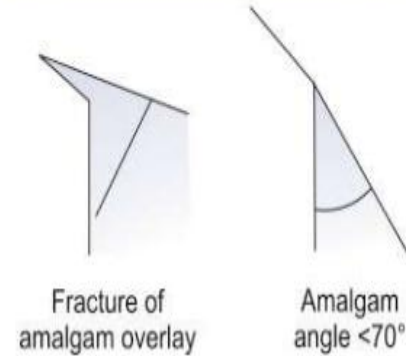
► **Pulpal floor**

- There should be flat pulpal floor of the preparation to avoid fracture of amalgam and the tooth. Curved floor for restoration acts as a wedge, which can result tooth fracture.

► **Cavosurface angle**

- The tooth and amalgam joint, i.e. cavosurface angle should be a butt joint especially where the masticatory stresses are present. If cavosurface angle is acute, enamel margins may fracture under load. But if cavosurface angle is obtuse, marginal amalgam may fracture under masticatory stresses.

Failure of amalgam margins



- ▶ **Unsupported enamel rods**

- ▶ Presence of unsupported enamel rods can result in fracture and thus secondary caries because of gap formation. They should be removed properly.

- ▶ **Proximal preparation**

- ▶ In proximal preparations, fracture of amalgam can occur because of inadequate width and depth of isthmus or insufficient proximal retention form.

- ▶ **Line angles**

- ▶ Fracture of amalgam restoration may occur because of sharp axio-pulpal line angle because of concentration of stresses in that area. So these angles should be rounded.

- ▶ **Incomplete removal**

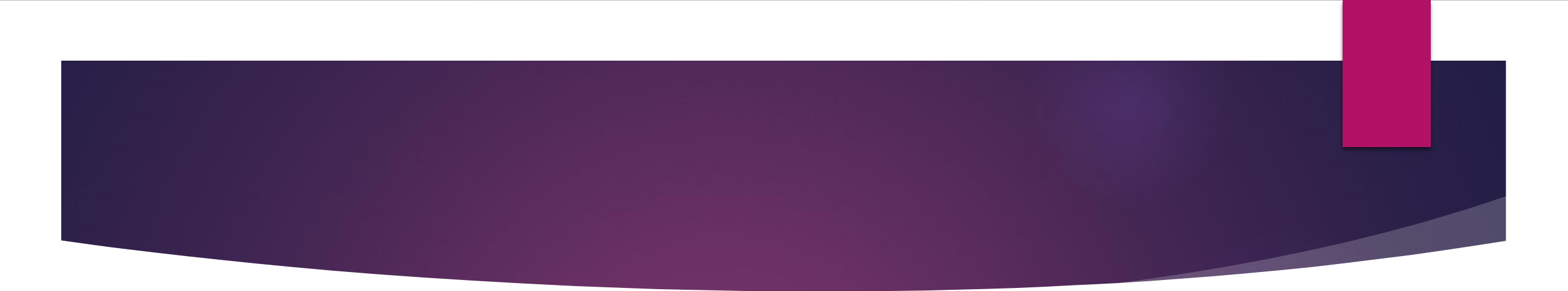
- ▶ Incomplete removal of the defective enamel before restoration can also result in fracture of the restoration under masticatory load

- ▶ **Postoperative pain**

- ▶ It may occur due to pulpal hyperemia. To avoid, this one should make use of effective cooling, sharp burs and intermittent cutting.

- ▶ **Additional retentive form**

- ▶ Wherever additional retentive forms and devices are used for additional retention, they should be prepared entirely in dentin



▶ **3. Defective Amalgam Manipulation**

▶ Defective amalgam manipulation may occur in the following forms.

▶ **Inappropriate condensation:**

▶ The purpose of condensation is to adapt amalgam to the preparation walls and floors and to express excess mercury from the amalgam. An ideal amalgam mix should have mercury content below 55 percent, this can be achieved during condensation.

Following points should be kept in mind while condensing amalgam

- ➔ Small increments of amalgam should be used to make sure proper condensation
- ➔ Use of adequate condensation pressure
- ➔ Avoid delay between trituration and condensation
- ➔ Amalgam mix should be used within three minutes of its trituration.

▶ **Incorrect mercury alloy ratio:**

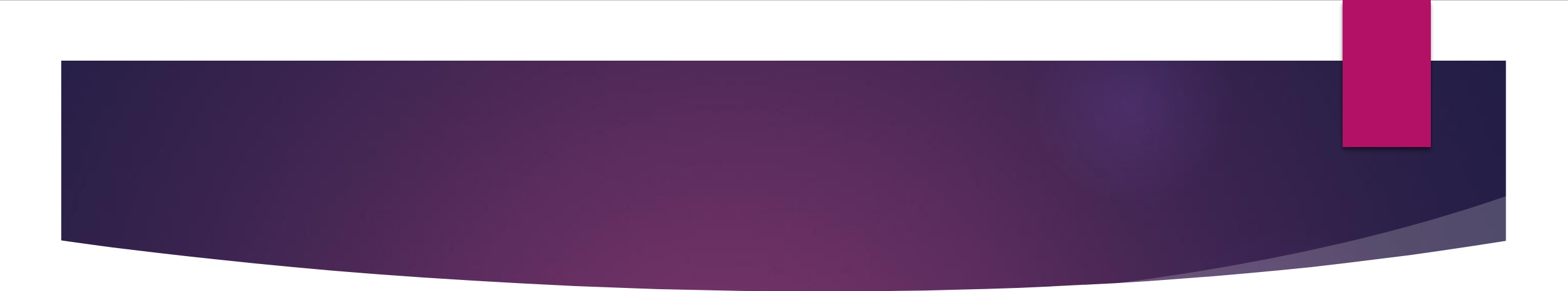
- ▶ Ideally in amalgam, the mercury content should be less than 55 percent. If it is more than 55 percent, there is decrease in strength of the amalgam. If large amount of mercury is used in trituration, the excess removal of mercury becomes difficult by mulling and condensation. Hence, before trituration, proper proportion of alloy and mercury should be used.

▶ **Contamination during manipulation:**

- ▶ While manipulation of amalgam if the amalgam gets contaminated with moisture, there occurs reduction in strength of amalgam.
- ▶ In zinc containing alloys, contamination with moisture causes delayed expansion resulting in pain, weakness at the margins, tarnish, and corrosion.

▶ **Faulty finishing and polishing:**

- ▶ Excessive heat production during polishing may result in pulpal trauma. Heavy pressure applied during polishing results in spur like overhangs, which fracture under mastication causing leaky margins and prone to secondary caries.



4. Defective Matrix Adaptation

- ▶ Proper matrix and retainer should be selected according to requirement. Matrix should be properly contoured according to the tooth type and stabilized using a wedge.
- ▶ If wedge is not used, excess material can go into gingiva and thus irritate the periodontium.
- ▶ Before condensation of amalgam, matrix should be properly made stable to avoid distortion of the restoration
- ▶ If matrix band is removed prematurely before the restoration is set, it may fracture the restoration.

5. Postrestorative Failures

Postrestorative pains: It can occur because of following reasons:

- ▶ Apical periodontitis or, fracture of the filling or tooth and pain
- ▶ In zinc containing alloys, delayed expansion can cause fracture of filling or tooth and pain.
- ▶ Production of galvanic currents.
- ▶ Extreme changes in temperature in oral cavity may cause pulpal hyperemia resulting in pain.
- ▶ Because of good thermal conductivity, insufficient pulp protection may give rise to pain. Thus it is advisable to use pulp protective materials beneath amalgam restoration.
- ▶ Premature fracture of restoration: If patient bites the restoration soon after its placement and before final setting of amalgam takes place, restoration may fracture. Therefore, postoperative instructions must be clearly explained to the patient.

Reading assignments

- ▶ Modification of class II design
- ▶ Cavity preparation for amalgam in class III , class V and class VI
- ▶ Recent advances in amalgam

References

- ▶ Text book of operative dentistry jaypee brothers 2nd edition
- ▶ sturdevant's art and science of operative dentistry